Supplementary Material: Lyapunov Resource Control for Correlated Multi-Agent Markov Learning

Jialin Zhuang and Vincent K. N. Lau

I. Proof of the Same-Stream Actual-Risk Enclosure

Consider the stationary common factor in the main paper, with independent private observation noise and a deterministic probe $d = (q_d, b_d, n_d)$ ending at $T = n_d b_d$. The probe sigma-field $\mathcal{H}_d$ contains its observations; spatial contrasts contain no additional information about the common factor. Under true instance j, Gaussian filtering gives

$$C_T \mid \mathcal{H}_d \sim \mathcal{N}(m_j, P_j),$$
$$m_j \sim \mathcal{N}(0, s_j^2), \quad s_j^2 = \theta_j - P_j. \tag{S1}$$

The posterior variance P_j is deterministic because the probe schedule is fixed before data. Starting from variance θ_j, the observation update is $P \leftarrow P/(1+q_d P)$; between observations it is $P \leftarrow \lambda^{2b_d} P + (1-\lambda^{2b_d})\theta_j$. The last update has no subsequent prediction.

Fix either downstream program a, including its budget, initialization, sample spacing and deterministic weights. All learning observations occur strictly after T. With $r_a = \lambda^{b_a}$ and the notation of main Eq. (38), its unconditional stationary risk is

$$J_{j,a} = d_a^2 + \theta_j \sum_{t,u=1}^{N_a} \omega_{a,t}\omega_{a,u} r_a^{|t-u|} + q_a^{-1} \sum_{t=1}^{N_a} \omega_{a,t}^2. \tag{S2}$$

Conditional means of future common states are $\lambda^{b_a t} m_j$. Their conditional covariance is their stationary covariance minus $s_j^2 \lambda^{b_a(t+u)}$. Private learning noise is independent of the probe. Therefore

$$\mathbb{E}_j[Z_a^2 \mid \mathcal{H}_d] = J_{j,a} + 2 d_a k_a m_j + k_a^2 (m_j^2 - s_j^2). \tag{S3}$$

This identity includes the posterior mean, which is unbounded across probe histories; no uniform conditional Gaussian bound is invoked.

Let E_j be the event that the decision selects the wrong frozen program, and let $p_j = \Pr_j(E_j) \le \delta$. Subtracting Eq. (S3) for the correct and wrong programs gives

$$\mathbb{E}_j Z_{\widehat{a}}^2 = J_{j,a_c} + p_j \Delta J_j + \mathbb{E}_j[\mathbf{1}_{E_j} U_j], \tag{S4}$$

where

$$U_j = 2(d_{a_w} k_{a_w} - d_{a_c} k_{a_c}) m_j + (k_{a_w}^2 - k_{a_c}^2)(m_j^2 - s_j^2). \tag{S5}$$

The centered Gaussian first and second moments are orthogonal, with variances s_j^2 and $2s_j^4$. Consequently $\mathbb{E}_j U_j = 0$ and $\mathbb{E}_j U_j^2 = \Gamma_j^2$ as defined in the main paper. Cauchy–Schwarz gives $|\mathbb{E}_j[\mathbf{1}_{E_j} U_j]| \le \sqrt{p_j}\Gamma_j \le \sqrt{\delta}\Gamma_j$. Bounding the signed term $p_j \Delta J_j$ and using nonnegative squared risk proves main Theorem 5.

II. Exact Weights of the Implemented Learners

Let $M_a = \min\{\lfloor B'_m/(h+q_a)\rfloor, \lfloor B'_e/b_a\rfloor\}$ be the number of scheduled observations after subtracting probe costs. All M_a observations are paid. The FIFO withholding convention leaves $N_a = (M_a - D)_+$ usable observations, not a free replacement budget. For $N_a = 0$ the reported parameter is one, so $d_a = 1$ and the weight vector is empty.

A. Running mean

The Layer-A output is $(1+\sum_{t=1}^{N_a} \bar{X}_t)/(N_a+1)$. Thus $d_a = 1/(N_a+1)$ and every $\omega_{a,t} = 1/(N_a+1)$. Main Eq. (38) then gives

$$k_a = \frac{r_a(1 - r_a^{N_a})}{(N_a+1)(1-r_a)}. \tag{S6}$$

B. Averaged FIFO affine learner

In Layer B, an increment $\eta(\bar{X}_t - \mu w_{t-1})$ is computed before enqueueing and is applied after D opportunities. The original implementation fixes $\mu = 1/2$ and $\eta = 0.1/[\mu(D+1)]$, starts from parameter one with an empty queue, and averages the applied parameters. Indexing these applied parameters as z_n gives

$$z_n = z_{n-1} - \eta\mu z_{n-D-1} + \eta \bar{X}_n, \quad z_{-D} = \cdots = z_0 = 1. \tag{S7}$$

Define $h_0 = 1$, $h_t = 0$ for $t < 0$, and $h_t = h_{t-1} - \eta\mu h_{t-D-1}$ for $t \geq 1$. For $N = N_a > 0$, the output $N^{-1}\sum_{n=1}^{N} z_n$ has exact coefficients

$$\omega_{a,t} = \frac{\eta}{N}\sum_{v=0}^{N-t} h_v, \quad d_a = 1 - \mu\sum_{t=1}^{N}\omega_{a,t}. \tag{S8}$$

To verify the deterministic term, constant input $\bar{X}_n = \mu$ keeps the initial history at one; linearity then yields the identity for d_a. Eq. (S8) represents the average of applied iterates; all scheduled observations remain charged.

III. Shrinking-Risk and Budget-Shift Interface

Suppose the finite action catalogue has $|\lambda| \leq 1 - \gamma < 1$, bounded D, and strictly stable affine recursions. For the registered step rule this stability follows directly from the characteristic polynomial $z^{D+1} - z^D + t$, $t = \eta\mu$. At small $t > 0$, the root at one moves inside the unit disk and the other roots remain inside. Writing a first unit-circle crossing as $z = e^{i\omega}$ yields $t = 2\sin(\omega/2)$ and the smallest positive crossing $\omega = \pi/(2D+1)$. Thus $0 < t < 2\sin[\pi/(4D+2)]$ is stable; the registered $t = 0.1/(D+1)$ lies strictly inside this interval. The impulse response consequently decays geometrically, and $\sum_{v\geq 0} h_v = 1/(\eta\mu)$.

The interior weights in Eq. (S8) are $1/(\mu N)$ up to a summable end correction: their difference from these constant weights has ℓ_1 norm $O(N^{-1})$ and maximum absolute value $O(N^{-1})$. The homogeneous transient is summable, so its average d_a is $O(N^{-1})$. The AR(1) covariance kernel has uniformly bounded absolute row sums. Substitution into Eq. (S2) therefore gives

$$J_{j,a} = \frac{1}{\mu^2 N_a}\left[\frac{1}{q_a} + \theta_j\frac{1+r_a}{1-r_a}\right] + O(N_a^{-2}). \tag{S9}$$

The running mean has the same expansion with $\mu = 1$. These constants are uniform over the finite catalogue and bounded instance class. On a budget ray, $N_a = s/c_a + O(v+D+1)$ with $c_a = \max\{(h+q_a)/\beta_m, b_a/\beta_e\}$ and fixed probe charge v. Hence $J_{j,a} = K_{j,a}/s + O((v+D+1)/s^2)$, with

$$K_{j,a} = \frac{c_a}{\mu^2}\left[\frac{1}{q_a} + \theta_j\frac{1+r_a}{1-r_a}\right]. \tag{S10}$$

Subtracting a fixed probe charge changes N_a by at most $O(v+D+1)$, giving the claimed budget-shift remainder. Moreover $k_a = O(N_a^{-1})$ because the weights are uniformly $O(N_a^{-1})$ and $\sum_{t\geq 1}|r_a|^t$ is bounded. Since $s_j^2 \leq \theta_j$, main Eq. (39) yields

$$\sqrt{\delta}\Gamma_j = O(\sqrt{\delta}/s^2). \tag{S11}$$

For a pair of frozen actions satisfying the main phase theorem's leading-gap and wrong-decision assumptions, adding this remainder to admission changes only its second-order class constants. The same necessary-information and sufficient-scale argument then applies to the corrected actual-risk rule. This is an averaged, additive Gaussian affine result; it does not remove the nonzero noise floor of a fixed-step unaveraged learner or the conditional-history requirements of the general Markov theorem.

TABLE I
Population enclosures for mean information-only minus learning-aware risk, rounded outward. All means use the listed complete stratum.

Layer	Stratum	Cells	Lower	Upper
A	All	1184	.0506	.0706
A	Intermediate	576	.0770	.1180
A	Original practical	446	.1607	.1952
A	Search-censored	32	.4879	.4880
B	All	608	.0079	.0223
B	Intermediate	288	-.0049	.0252
B	Original practical	222	.0562	.0773
B	Search-censored	32	.1957	.1958

IV. Complete Existing-Evidence Reanalysis

The calculation uses all registered EXP-016B budget points, both regimes and all eight policies: 14,336 policy cells, each with its original 192-seed empirical mean. Layer A contains 104 scenario families and Layer B its original 56, including eight search-censored families in each layer. For fixed policies Eq. (S2) is exact. For probe policies main Eq. (40) supplies population-risk bounds. When information-only and learning-aware policies execute the same frozen program, their paired difference is identically zero, not the difference of two independent interval bounds. Otherwise subtract their one-sided bounds.

Below identification, all 208 Layer-A and 112 Layer-B comparisons are identical programs; above finite value thresholds the corresponding counts are 384 and 192. In the original practical subset, 323/446 Layer-A and 176/222 Layer-B cells have strictly positive improvement lower bounds, while 101 and 33 have strictly negative upper bounds; the remaining 22 and 13 are inconclusive. These include regime-specific losses even when the stratum-average improvement is positive. No new admission rule was executed and no old gate was replaced. The old probe-length routine used the sufficient exponent $\log(0.5/0.025)$, giving an equal-prior average-error target of 0.025 rather than a directional guarantee of 0.025. The reanalysis uses each probe's exact affinity $\exp(-I_B)$ as a valid bound on either directional error; over all 104 frozen probe designs these bounds range from 0.0352271 to 0.0497465. Failure of this sufficient directional certificate does not establish that the actual

directional errors exceed 0.025. No outcomes or probe lengths are changed to obtain a smaller error bound. This outcome-known reanalysis gives population enclosures, not new confidence intervals for empirical means.

V. Complete Threshold-Sandwich Scale Calculation

For the independently initialized Gaussian running mean, let $c_a = \max\{(h+q)/\beta_m, b/\beta_e\}$ and $r = \lambda^b$. The exact horizon is $N_a = s/c_a + O(1)$, with bounded terminal charges and rounding. Main Eq. (8) and Eq. (13) give $K_j(a) = c_a\{q^{-1}+\theta_j(1+r)/(1-r)\}$ and a uniform $O(s^{-2})$ remainder. Removing probe charge v changes N_a by $O(v+1)$ and hence risk by $O((v+1)/s^2)$ for $s \geq s_0(v+1)$; compact persistence and the finite catalogue make these constants uniform. Thus the phase-theorem risk conditions have a concrete Gaussian realization, separately from the constant-step affine certificate. Write $d_\delta = \mathrm{kl}(1-\delta,\delta)$, $r_0 = 1-\gamma < 1$, and $Q = \max \mathcal{Q}$. Under either true instance, predictable spacing gives $\|C_{S_k}\|_{L^2} \leq r_0\|C_{S_{k-1}}\|_{L^2} + \sqrt{\theta_+}$, hence a uniform bound $M_C = \sqrt{\theta_+}/(1-r_0)$. The normalized observation $Y_k/\sqrt{q_k}$ has L^2 norm at most M_C+1. For either candidate filter, write its posterior mean as $(1-t_{j,k})m^-_{j,k} + t_{j,k}Y_k/\sqrt{q_k}$, with predictable $0 \leq t_{j,k} \leq 1$. Prediction multiplies this mean by a factor of magnitude at most r_0; the triangle inequality bounds both predictive means uniformly in L^2 by $M_F = r_0(M_C+1)/(1-r_0)$, starting from zero. Both predictive variances are at most θ_+, so $1 \leq V_{j,k} \leq V_+ = 1 + Q\theta_+$. Main Eq. (26) gives the expected, rather than pointwise, bound $\mathbb{E}_j\mathcal{I}_{j,1-j}(Z_k, A_k) \leq i_+$, where $i_+ = (\log V_+ + V_+ + 4QM_F^2)/2$ is finite. Extend a stopped rule predictably after stopping; nonnegativity and its deterministic budget bound $n_{\max}(s) \leq c_B s$ imply $\mathbb{E}_j \sum_{k\leq\tau}\mathcal{I}_{j,1-j} \leq i_+ n_{\max}(s)$. Together with main Eq. (30), this proves $B_N \geq c_N d_\delta$. The fixed probe has information at least $i_- > 0$, so $n_\delta = \lceil i_-^{-1}\log(1/\delta)\rceil$ observations suffice and cost at most $c_p n_\delta + D_{\max}$. Continuity of $\log(1/\delta)/d_\delta$ gives $n_\delta \leq c_0 + c_1 d_\delta \leq c_0 + c_2 B_N$. Write $u_\delta \leq c_p n_\delta + D_{\max}$ for the full probe charge. The assumed expansion and budget-shift inequality give a high-instance improvement over the full-budget fallback certificate of at least $g_{\min}/s - \{2M + L(u_\delta+1)\}/s^2 - \delta w_+/s$. It is strictly positive for $s > \{4M+2L(u_\delta+1)\}/g_{\min}$, also requiring $s \geq s_0(u_\delta+1)$. On instance zero, the assumed coefficient ordering gives $\epsilon_{\mathrm{safe}} \leq \{2M+L(u_\delta+1)\}/s^2 + \delta_0 w_+/s$. Thus $s \geq \sqrt{2\{2M+L(u_\delta+1)\}/\bar{\epsilon}} + 2\delta_0 w_+/\bar{\epsilon}$ suffices. Using $2\sqrt{xy} \leq x+y$, all three requirements are bounded by a class-constant multiple of $n_\delta + B_{\mathrm{oracle}} + B_\epsilon + 1$. Taking the maximum proves the upper bound in main Eq. (46); its lower bound follows because every admitted δ-correct probe obeys both information constraints.

VI. Robust-Certificate Sensitivity Details

This section supplies the proof of main Corollary 3. Write $\delta^+ = C_q(r_q^+)^b$. On the coverage event, $\delta^+ \geq \delta$ and monotonicity on $\mu - 2L\delta^+ > 0$ gives $a^+ \geq a$ and $\beta^+ \geq \beta$. Evaluating the delayed Young inequality at the shared $\lambda^+ = \sqrt{h/a^+}$ proves the upper recursion; separately optimized forcing terms need not be monotone. For $\Delta\delta = \delta^+ - \delta$, $\Delta K = K_q^+ - K_q$, and $\Delta\Omega = \Omega_q^+ - \Omega_q$,

$$\begin{aligned}
\Delta a &= 2\eta^2\Delta K + (2\eta L + 4\eta^2L^2)\Delta\delta,\\
\Delta\beta &= 2\eta^2\Delta\Omega + 4\eta^2G^2\Delta\delta\\
&\quad + 4\eta G^2\left[\frac{(\delta^+)^2}{\mu - 2L\delta^+} - \frac{\delta^2}{\mu-2L\delta}\right],\\
\Delta c &= (\sqrt{a^+}+\sqrt{h})^2 - (\sqrt{a}+\sqrt{h})^2. \qquad \text{(S12)}
\end{aligned}$$

For $h > 0$,

$$\begin{aligned}
\Delta d &\leq (1+\sqrt{h/a^+})\Delta\beta + \sqrt{h}\left(\frac{1}{\sqrt{a}} - \frac{1}{\sqrt{a^+}}\right)\beta\\
&\quad + \frac{\sqrt{a^+}-\sqrt{a}}{\sqrt{h}}g, \qquad \text{(S13)}
\end{aligned}$$

and $\Delta d = \Delta\beta$ when $h = 0$. For $U(c,d) = d/(1-c) + c^n(R_0 - d/(1-c))_+$, $|\partial_d U| \leq (1-\bar{c})^{-1}$ and $|\partial_c U| \leq nR_0 + \bar{d}(1-\bar{c})^{-2}$. The mean-value bound proves main Eq. (48); applying it to the upper- and exact-score minimizers gives $U_{\hat{a}} \leq U_{a^\star} + 2\max_a \varepsilon_a$.

VII. Stochastic-Delay Interpretation

The controlled stochastic delay differential equation associated with one fixed resource action is

$$dE(t) = -A_q E(t-\tau(t))dt + \sqrt{\eta}\,\Sigma_q^{1/2}(E_t)dW(t). \qquad \text{(S14)}$$

Here $\Sigma_q(E_t)$ retains both additive innovation covariance and the state-dependent quadratic variation of random temporal-difference Jacobians. For $V(E) = \|E\|^2$, Itô's formula gives

$$\mathcal{L}_a V(E_t) = -2E(t)^\top A_q E(t-\tau(t)) + \eta\,\mathrm{tr}\,\Sigma_q(E_t). \qquad \text{(S15)}$$

Participation controls spatial covariance, spacing controls mixing, and η controls their scale; the discrete theorem retains finite steps, integer delays and rounding.

VIII. Constructing a Certificate from a Known Finite Model

Let Z be the finite augmented state containing all sources needed for the retained joint TD tuple. For each participation q, suppose its kernel M_q and stationary law π_q are known, and the marginal target policy and reward law are preserved. Tabulate $H_i(z)$ and $c_i(z)$, put $A = \mathbb{E}_{\pi_q}H_i$, $c = \mathbb{E}_{\pi_q}c_i$, and solve $Aw^\star = c$. Check that these means are common to all actors and

all q, and that $\mu = \lambda_{\min}((A + A^\top)/2) > 0$. With $\xi_i(z) = c_i(z) - H_i(z)w^\star$, compute

$$L = \max_{i,z} \|H_i(z)\|_2, \qquad G = \max_{i,z} \|\xi_i(z)\|,$$
$$K_q = \lambda_{\max}\left(\sum_z \pi_q(z)\bar{H}_q(z)^\top \bar{H}_q(z)\right),$$
$$\Omega_q = \sum_z \pi_q(z)\|\bar{\xi}_q(z)\|^2,$$
$$\beta_q(g) = \max_z \frac{1}{2}\sum_{z'} |M_q^g(z, z') - \pi_q(z')|. \tag{S16}$$

Here bars denote actor averages, not averages of squared norms. The integer g is the number of transitions of the augmented tuple chain between the conditioning state and the retained tuple; it must be derived from the declared physical schedule, not identified with a state lag or drain charge.

Proposition 1 (Constructive finite-model interface)**.** *If M_q is irreducible and aperiodic and the above common-mean and positivity checks pass, Assumption 1's boundedness, centering and conditional mixing inputs hold with Eq.* (S16) *and $\delta = \beta_q(g)$. For every action whose resulting μ_δ, a_δ and $c_{\mathrm{aff}} < 1$ pass the main paper's shield, the affine certificate is valid from any declared initial augmented state.*

Proof. The finite maxima bound every sample and stationarity gives $\mathbb{E}_{\pi_q}\xi_i = 0$. Conditional on any earlier history, the Markov property expresses the next tuple law as a mixture of rows of M_q^g. Convexity of total variation bounds its distance from π_q by $\beta_q(g)$, uniformly over that history. The drift proof uses only this total-variation distance, so the geometric envelope can be replaced by this exact finite-gap value. The stated positivity and contraction checks are precisely the remaining hypotheses of the affine bound. □

This is a construction, not a statistical estimator of an unknown kernel. A dense joint-state table may grow exponentially with participation; its offline construction cost is separate from the online scalar score. In the seven-state shared/private-source model, a product coupling gives the cheaper bound $\beta_q(g) \le \min\{1, (q+1)\beta_{\mathrm{pair}}(g)\}$, where $\beta_{\mathrm{pair}}(g) = \max_x \|M_{\mathrm{pair}}^g(x, \cdot) - \pi_{\mathrm{pair}}\|_{\mathrm{TV}}$. This uses the actual consecutive-pair chain and independent latent source chains; fresh selector coins are integrated by data processing. The existing implementation computes these pair kernels and the same-time aggregate moments directly. For a moving target policy, a frozen-kernel table does not certify mixed histories: either supply uniform bounds for the nonhomogeneous kernel products or reset the learner and exclude all pre-reset target updates.

IX. Executable Actual-Risk Gate and Saved-Result Reconstruction

For each fixed candidate probe, Gaussian filtering gives P_j, the formulas above give $(J_{j,a}, d_a, k_a)$, and the probe likelihood laws give a directional error bound δ_j under each hypothesis. The exact Gaussian affinity is one valid choice: a zero-threshold likelihood-ratio decision obeys $\Pr_j(\mathrm{error}) \le \int \sqrt{p_0 p_1}$ in each direction. An equal-prior average-error bound is not interchangeable with this directional bound. Compute both hypothetical-instance quantities before probing; the actual unknown instance is not a gate input.

Proposition 2 (Actual-risk admission)**.** *Fix the two hypotheses, noise law, learner initialization and deterministic linear program weights as in Theorem 5. Given a finite public probe catalogue, choose a feasible design satisfying Eq.* (41) *before any probe observations, with deterministic tie breaking. After its likelihood decision, execute its frozen selected program on the continuing stream. Then $\mathbb{E}_1 Z^2 < F_1$ and $\mathbb{E}_0 Z^2 \le F_0 + \bar{\epsilon}$. If none passes, no probe is executed and the actual risk is exactly F_j.*

Proof. For each fixed design Theorem 5 gives $\mathbb{E}_j Z^2 \le U_j(d)$. Pre-observation deterministic choice leaves its fixed-design hypothesis intact. The two gate inequalities imply the conclusions. In the rejection branch the stationary full-budget commitment defines F_j. □

Thus the procedure is: price the probe, compute the remaining horizons and exact weights for both branch programs, compute U_0, U_1, F_0, F_1, apply the two inequalities, and only then acquire data. An observation-adaptive choice of probe duration or of program weights is not covered by this fixed-design argument.

We reconstruct this gate on the existing EXP-016B outcomes, using its single registered information-only probe candidate and the original scenario-specific allowances 0.1 or 0.2. All 896 public layer/scenario/budget designs and both regimes are retained, including search-censored families. The corrected gate admits 480 designs, versus 505 without the carryover term; it removes 25 admissions. The outcome-known zero-allowance stress case admits none under either rule. It is retained separately: a paid probe can reduce the low-instance learning horizon even with perfect identification.

The gate is a deterministic function of public design quantities, so an admitted design takes the saved information-only outcome and a rejected design takes the saved no-probe fallback outcome. The simulator's potential observations are policy independent and shared across these branches; this reconstructs their empirical means without generating or resampling a trajectory. Table II is therefore a retrospective policy comparison,

TABLE II
SAVED-RESULT GATE RECONSTRUCTION: MEAN SQUARED RISK OVER ALL BUDGET/REGIME CELLS, WITHOUT NEW TRAJECTORIES. EACH ORIGINAL CELL MEAN CONTAINS ALL 192 SAVED SEEDS.

	A high	A low	B high	B low
Full-budget fallback	.63455	.09201	.82703	.54369
Information-only	.54814	.24062	.76422	.62515
Original learning-aware	.57637	.09304	.79121	.55825
No-carryover gate	.52548	.09488	.74204	.56252
Corrected gate	.52989	.09393	.74242	.56068

not fresh execution or independent confirmation. Against fallback, the corrected gate retains $13/592$ and $4/304$ empirically adverse high-instance cells in A and B; finite-sample cell means need not satisfy the population ordering. Its low-instance mean risks exceed fallback by $.00192$ and $.01699$, respectively, within the declared nonzero safety allowances, not a claim of zero-harm safety. The no-carryover ablation has slightly lower high-instance means but higher low-instance means; the correction trades admissions for a valid population-risk guarantee, not uniformly better observed risk. All eight original policies, per-cell costs, both replay variants and their complete cell tables are preserved. The new comparison was specified after earlier outcomes were known and supplies no new confidence interval.

X. NEAREST-WORK ASSUMPTIONS AND OBJECTIVES

The comparison uses the accessible full preprints, with theorem and proof locators recorded in the accompanying source-reading matrix; it is not an exhaustive priority claim. DASA [25], Assumption 3 and Theorem 1, combines independent agent chains with median-based update filtering to obtain average-delay dependence and parallel speedup. Its Lemma 2 uses cross-agent independence in the noise cross terms. Our bounded deterministic delay condition is narrower; our added object is covariance-dependent paid learning risk, not a stronger delay rate.

Perazzone et al. [20], Theorem 1 and Eqs. (12)–(23), link inverse participation probabilities to a nonconvex convergence bound, then optimize a time-average convergence/communication surrogate with power queues. Their solver and instantaneous-channel formulation address a different operational problem; our finite-budget probe debit and same-stream decision-event correction do not follow from that time-average objective. Yu et al. [26], Sections III–IV and Propositions 12–14, use an SDDE/arrival-process approximation to optimize worker activation, including a finite-service queue. That queue model and its delay law are not established by our fixed-delay experiment.

Saad et al. [32], Assumption 1, Protocol 1 and Theorems 4.1/5.1, adapt joint arm queries to unknown covariance with independent rounds and a fixed-confidence identification objective. Our dependence hypotheses are supplied, rather than learned nonparametrically; our comparison instead prices identification through the subsequent learner. Li et al. [35], Theorem 4.12, remove unknown mixing/eigenvalue inputs from a regularized TD step schedule. That reduces tuning requirements that our general confidence interface still has, but does not construct a calibrated multi-action terminal-risk shield. Li et al. [36], Assumptions 1–7 and Theorem 1, give shrinking-tube reliability for projected adaptive-kernel SA using uniform drift/minorization and kernel continuity. Their mixed-history result is broader than a frozen-kernel table; our Gaussian decision correction and paid finite horizon are different outputs, not a replacement for their regularity assumptions. Reference numbers in this supplement refer to the main paper's bibliography.

XI. COMPLETE TRACKING PROOF

Set $R_\star = d/(1-c)$ and $\Gamma_r = (Z_r - R_\star)_+$. The explicitly assumed uniform-history drift and the positive-part replacement argument give

$$\begin{aligned} \widetilde{Z}_{r+1} &\le R_\star + c\Gamma_r, \\ \Gamma_{r+1} &\le (1+\chi)c\Gamma_r + \chi R_\star + (1+\chi^{-1})v^2. \end{aligned} \tag{S17}$$

The second line follows by applying Young's inequality to the target shift and subtracting $R_\star$. Iteration and $\Gamma_0 \le Z_0$ prove Eq. (20) whenever $\bar{c} = (1+\chi)c < 1$. Finally, $\widehat{A}_w - A^\star = \{\gamma\phi(s') - \phi(s)\}^\top (w - w_n^\star)$ and $\|\gamma\phi(s') - \phi(s)\| \le (1+\gamma)\Phi$; Cauchy–Schwarz proves Eq. (21).

XII. COMPLETE OCCUPATION-MEASURE PROOF

Let $\Delta_j(z,a) \ge 0$ be the one-step opportunity loss relative to the instance-j certified oracle. For a stopped predictable policy, define

$$\nu_j(G,a) = \mathbb{E}_j^\pi \sum_{k \le \tau} \mathbf{1}\{Z_k \in G, A_k = a\}. \tag{S18}$$

Every δ-correct policy induces an occupation measure satisfying the controlled Kalman flow constraints,

$$\begin{aligned} \int \mathcal{I}_{j,1-j}(z,a)\nu_j(dz,da) &\ge \mathrm{kl}(1-\delta,\delta), \\ \int (h + q(a))\nu_j(dz,da) &\le B_{\mathrm{msg}}, \\ \int b(a)\nu_j(dz,da) + \ell\mathbb{P}_j(\tau > 0) &\le B_{\mathrm{env}}. \end{aligned} \tag{S19}$$

Let T_a be the belief transition and ζ_j the terminal belief law. For every bounded measurable f, predictability and stopped telescoping give

$$\int f(z)\zeta_j(dz) - f(z_1) = \int \Big[\int f(z')T_a(z, dz') - f(z)\Big]\,\nu_j(dz, da). \tag{S20}$$

Moreover,

$$\mathbb{E}_j^{\pi} \sum_{k\le\tau} \Delta_j(Z_k, A_k) = \int \Delta_j(z, a)\nu_j(dz, da). \tag{S21}$$

Eq. (29), Eq. (6), and Eq. (S20) place every admissible occupation measure in the controlled feasible set. Taking the infimum of its exact loss over that larger set proves Proposition 4.

XIII. Complete Cooperative Policy-Learning Transfer Results

Held-out MAPPO transfer studies match message/environment budgets and charge a disjoint critic-residual probe that selects participation without evaluation return. Both studies use eight fresh seeds per dependence regime and compare against fixed $q \in \{1, 2, 4, 8\}$. We compare regime-wise fixed envelopes and the strongest single fixed action for an unknown regime.

In PettingZoo `simple_spread_v2`, the controller commits to $q = 8$ in all eight independent-stream runs and $q = 1$ in all eight shared-stream runs. It lies within 0.357 and 0.889 percent of the corresponding regime-wise fixed envelopes. Against the strongest single fixed action, $q = 8$, the mean of the regime–seed normalized AUC gains is 2.319 percent; normalization precedes averaging.

In MaMuJoCo `HalfCheetah-v2/2x3`, the task-specific prespecified catalogue is $\{2, 8\}$, while evaluation still includes all four fixed actions. The controller selects $q = 8$ in all eight independent-stream runs and $q = 2$ in all eight shared-stream runs. Relative to the corresponding regime-wise fixed envelopes, its AUC changes are -0.542 and -0.997 percent, both within the prespecified two-percent noninferiority margin. When the regime is unknown, the strongest single fixed action is $q = 8$. Forming each seed's equal-regime mixture before normalization gives a mean paired AUC gain of 8.341 percent, a one-sided 95 percent lower bound of 1.107 percent, seven of eight positive pairs ($p = 0.0352$), and a final-return gain of 4.637 percent.

Fig. 1 shows the unsmoothed deterministic team-return curves against the maximum charged dual-budget fraction. Probes consume at most 0.768 percent of messages; selection overhead is below 10^{-6} of training time.

XIV. Direct Fixed-Model Validation of the Actual-Risk Gate

A. Models, learners, and paid protocol

This study executes the actual-risk gate prospectively on new seeds, rather than reconstructing decisions from the earlier saved study. The 16 public families cross $Q \in \{8, 32\}$, high-instance variance $\theta_1 \in \{0.5, 2\}$, common-factor persistence $\lambda \in \{0.2, 0.8\}$, and withheld-update delay $D \in \{0, 8\}$. The low-instance variance is $\theta_0 = 0.05$, private variance is one, and message overhead is $h = 8$. Each family is evaluated under both true instances, each learner, and scales $s \in \{32, 128, 512, 2048\}$, with $B_{\mathrm{msg}} = (h+Q)s$ and $B_{\mathrm{env}} = 4s$. The 128 new seeds are $20571001, \ldots, 20571128$, screened against the historical seed registries before execution. All six policies share policy-independent potential observations within each seed, family, true instance, and learner, including across budget scales. They do not share potential observations across learners.

The common observation component is stationary Gaussian autoregressive noise, with coefficient λ and marginal variance θ_j; each actor has independent unit-variance Gaussian private noise. An average over the first q actors therefore has private variance $1/q$. Both learners start from one with a constant-one initial history. The running mean uses its exact averaging weights; the averaged delayed affine learner uses $\mu = 0.5$ and $\eta = 0.2/(D + 1)$ with the first-in, first-out recurrence analyzed above. Every scheduled block is paid, including the last D withheld updates: M scheduled updates give $N = \max\{0, M - D\}$ learner updates. There is no free startup or independent learning reset after a probe. The measured endpoint is squared parameter error at the exact zero target, not an unobserved within-run convergence curve.

Before data, the fixed probe uses $q_d = Q$, $b_d = 1$, and the smallest integer $n_d \in \{1, \ldots, 4096\}$ whose exact Gaussian affinity is at most 0.025. The log-likelihood threshold is zero, with ties assigned to the low instance. An infeasible or unavailable probe falls back to the full-budget all-agent program. After charging the probe, the two commitment programs minimize exact stationary risk under their respective supplied hypotheses over powers-of-two $q \le Q$ and $b \in \{1, 2, 4, 8\}$, breaking ties lexicographically in (q, b). Both programs are fixed before observing the probe; the true instance is not an input to any deployment decision. Writing F_j for the full-budget all-agent population risk and U_j for the same-stream population upper bound, the gate requires $U_1 < F_1$ and $U_0 \le F_0 + 0.1$. The no-carryover ablation deletes only the $\sqrt{\delta_j}\Gamma_j$ contribution from that admission calculation. The other controls are information-only

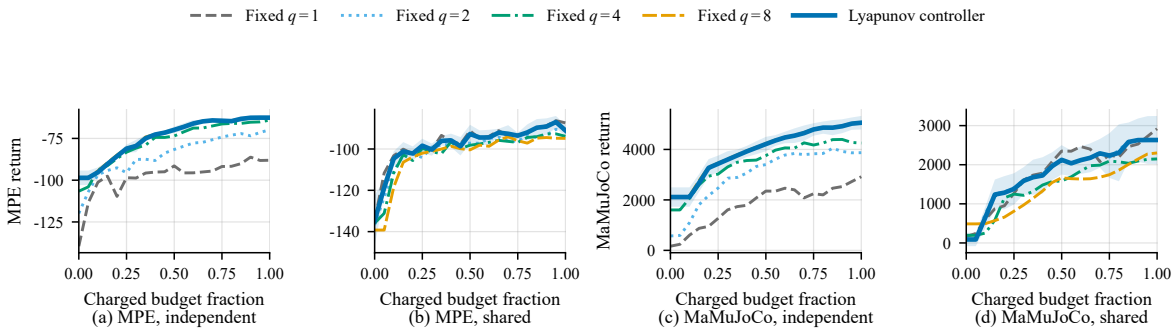


Fig. 1. Fully charged MAPPO team-return learning curves on two cooperative environments with fixed $q \in \{1, 2, 4, 8\}$ and the Lyapunov probe-then-commit controller (eight held-out seeds per regime; shaded bands are controller 95 percent confidence intervals). The controller selects $q = 8$ for independent streams in both environments, $q = 1$ for shared MPE streams, and $q = 2$ for shared MaMuJoCo streams.

probing, all-agent fallback, fixed $(q, b) = (2, 1)$, and a known-instance oracle used only for evaluation.

The complete grid contains $16 \times 2 \times 2 \times 4 \times 6 \times 128 = 196608$ trajectories. Every trajectory is regenerated in a full semantic replay, including its paid ledgers and decisions. Two analyses reproduce all seven output tables byte for byte. No failed, infeasible, adverse, or partial-cost record is removed.

B. All macro comparisons and counterevidence

For each learner, instance, and scale, we average the paired fallback-minus-policy squared-risk difference across all 16 families within each seed. Table III reports all 16 corrected-gate macro means and descriptive percentile intervals from 2,000 paired-seed bootstrap resamples. These are cellwise descriptive intervals, not a familywise significance or equivalence claim.

All eight high-instance macro intervals are positive, as are seven of eight low-instance intervals. The exception is the low-instance running mean at scale 32, where probing can spend resources without compensating risk reduction. At the finer family level, 23 of 256 corrected-gate mean improvements are negative (four high and nineteen low); 174 intervals are strictly positive, nine strictly negative, and 73 include zero or are exactly zero. The permitted low-instance population-risk increase is 0.1, so this adverse result does not contradict a zero-harm condition: no such condition was imposed.

The corrected gate admits probing on 113 of 128 public designs, exactly as does the no-carryover gate. Their probe-use indicators, actions, and risks coincide on all 32,768 paired policy rows. The selection-dependence term remains necessary to the stated bound, but its incremental empirical decision benefit is unresolved on this grid. Small differences between separately resampled bootstrap endpoints for the two identical policies do not constitute performance differences. The prespecified zero-allowance admission diagnostic admits 100 of 128 designs; it uses saved branch calculations, not additional trajectories.

Table IV retains all six policies' pooled risks and paid costs. Pooling gives equal weight to heterogeneous families, instances, learners, and scales, so it is descriptive rather than a universal ranking or a comparison at identical realized costs. Fixed $(q, b) = (2, 1)$ has lower pooled risk and environment cost than the corrected gate, whereas the gate sends fewer messages. The oracle is an instance-aware reference and is not deployable without the unknown label.

C. Probe validity and the scope of the enclosures

There are 8,192 distinct deduplicated potential probe decisions and thirteen observed classification errors, an empirical fraction of 0.001587. The exact affinity bounds range from 0.021139 to 0.024380, each below 0.025 under both supplied hypotheses. This new fixed-probe construction does not retroactively validate the earlier probe's nominal 0.025 directional interpretation, which bounded equal-prior average error instead. The historical reanalysis continues to use its larger exact affinities.

The risk enclosures concern population expectations under known binary Gaussian hypotheses and deterministic linear programs. In 123 of 256 corrected-gate cells, the finite sample mean exceeds the corresponding population upper bound; such an exceedance is not by itself a population-bound violation. In particular, fallback risk is an exact model expectation, not a simultaneous upper confidence limit for noisy sample means. The study therefore does not establish samplewise coverage, calibration for unknown kernels, or a general neural policy-learning guarantee.

Figs. 2–5 show every policy against budget scale, actual paid messages, and actual paid environment units, separately for each learner and instance. Each point

TABLE III
ALL CORRECTED-GATE MACRO RISK IMPROVEMENTS OVER ALL-AGENT FALLBACK. POSITIVE VALUES FAVOR THE GATE; BRACKETS GIVE DESCRIPTIVE 95 PERCENT PAIRED-BOOTSTRAP INTERVALS.

Learner	Scale	High instance: mean [interval]	Low instance: mean [interval]
Running mean	32	0.07820 $[0.06227, 0.09303]$	-0.002290 $[-0.003226, -0.001357]$
	128	0.03452 $[0.03051, 0.03840]$	0.000763 $[0.000530, 0.000986]$
	512	0.009436 $[0.008400, 0.010491]$	0.000373 $[0.000324, 0.000422]$
	2048	0.002268 $[0.001998, 0.002550]$	0.0000860 $[0.0000736, 0.0000985]$
Delayed affine	32	0.10001 $[0.08038, 0.11922]$	0.06114 $[0.05777, 0.06469]$
	128	0.14696 $[0.13225, 0.16261]$	0.07750 $[0.07538, 0.07955]$
	512	0.04006 $[0.03541, 0.04538]$	0.010206 $[0.009663, 0.010712]$
	2048	0.008698 $[0.007723, 0.009715]$	0.000914 $[0.000838, 0.000993]$

TABLE IV
DESCRIPTIVE POOLED MEANS OVER ALL 256 CELLS FOR EACH POLICY; EVERY CELL HAS 128 SEEDS.

Policy	Squared risk	Paid messages	Paid environment	Probe fraction
All agents	0.114864	19040.000	680.000	0
Fixed $(2, 1)$	0.062931	19037.500	1903.750	0
Information only	0.099064	18509.108	2467.163	0.937500
Oracle (evaluation only)	0.051335	18508.477	2509.625	0
Without carryover	0.079311	18509.341	2463.203	0.882813
Corrected gate	0.079311	18509.341	2463.203	0.882813

averages all 16 families and all 128 seeds; connecting points does not imply a within-trajectory learning curve or equal realized cost.

XV. ADDITIONAL RESOURCE-ACTION AND PROBE-VALUE PLOTS

Fig. 6 retains the complete action plot for the joint-controller study in the main paper. Fig. 7 retains its controlled probe-value comparisons; no data, caption or graphical content has changed.

A. Retained confidence-stress and reproducibility details

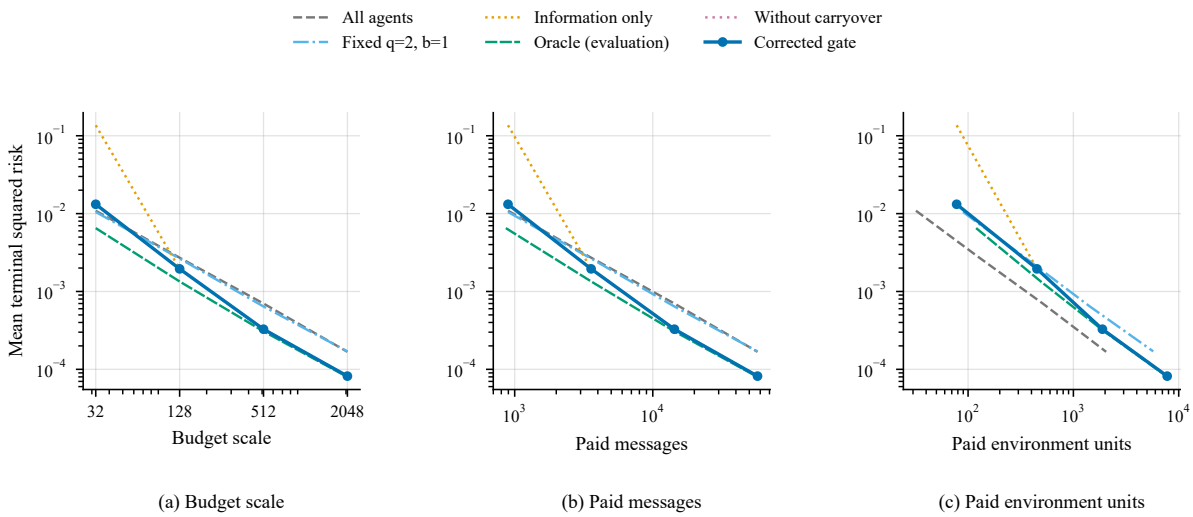
The Markov temporal-difference task has seven states on a ring, a bounded sinusoidal reward, four stationary-distribution-orthonormal features, and an exact projected fixed point. Cross-agent dependence is induced by shared and private Markov sources while preserving every agent's marginal transition law. The policy-learning studies use MAPPO with parameter-shared actors and a centralized critic [47]. They cover PettingZoo's multi-agent particle environment `simple_spread_v2` [48] and the multi-agent MuJoCo (MaMuJoCo) `HalfCheetah-v2/2x3` benchmark [49]. Bootstrap comparisons resample paired seeds and report percentile intervals for the ratio of geometric-mean risks. The confidence-interface study uses 64 disjoint confirmation seeds and 157,440 checkpoint rows. The moderate stress inflates K_q and Ω_q by 10 percent and weakens the certified mixing log-rate by 0.05; the strong stress uses 25 percent and 0.10. Both shields retain the nominal action in all twelve cells, all 24 stress-by-cell one-sided 99 percent upper means remain below the robust certificate, and neither shield changes empirical terminal risk. Controlled statistics/curves reproduce byte for byte; robustness uses eight development and 64 disjoint confirmation seeds. Each policy-learning study has 80 finite, exactly budgeted runs spanning both regimes and all policies; all 160 runs reproduce aggregate curves byte for byte. These ablations quantify the terminal parameter-risk gain from the finite-horizon objective; the joint-action study isolates the response to correlation, spacing, and delay.

XVI. SIGNED CONTINUING-STREAM RISK AND FIXED-PROGRAM SEPARATION

The design-utility formulation is classical [37]; selection dependence and perfect-information value are also established objects [38]. Here these tools are specialized to the paid, continuing-stream learner in main Eq. (38). The distinctions below are the sign of its actual-risk transfer, explicit one-sided constants, and a full-budget comparator, not new general association, truncation, or decision-theory principles.

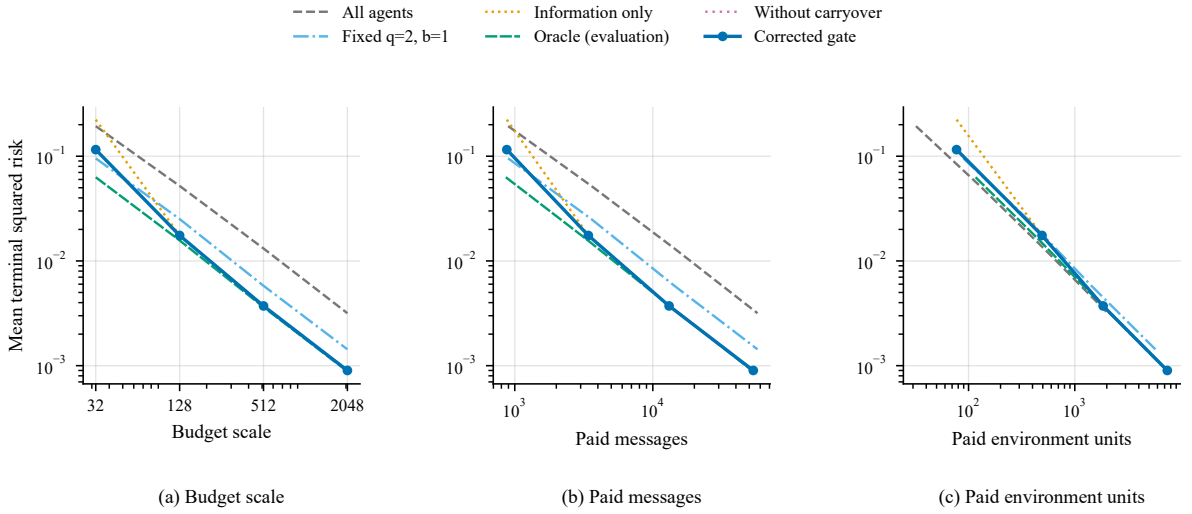
A. Exact signed transfer for a fixed likelihood decision

Freeze a_0, a_1, their remaining-budget horizons, and all weights before a deterministic probe. Its vector Y has laws $\mathcal{N}(0, \Sigma_j)$, where $0 \prec \Sigma_0 \preceq \Sigma_1$. Use a fixed likelihood threshold and a fixed data-independent tie rule, selecting one for the outside event $Y^\top(\Sigma_0^{-1} - \Sigma_1^{-1})Y > c$. Let E_j be the wrong-decision event, $p_j = \Pr_j(E_j)$, $\Delta J_j = J_{j,a_{1-j}} - J_{j,a_j}$, and $\Delta_k = k_{a_1}^2 - k_{a_0}^2$. The



(a) Budget scale (b) Paid messages (c) Paid environment units

Fig. 2. Direct validation, running mean and low instance: all-policy risk versus allocated scale and actual paid resources. Corrected and no-carryover curves coincide.



(a) Budget scale (b) Paid messages (c) Paid environment units

Fig. 3. Direct validation, running mean and high instance, with the same complete-family averaging and all six policies.

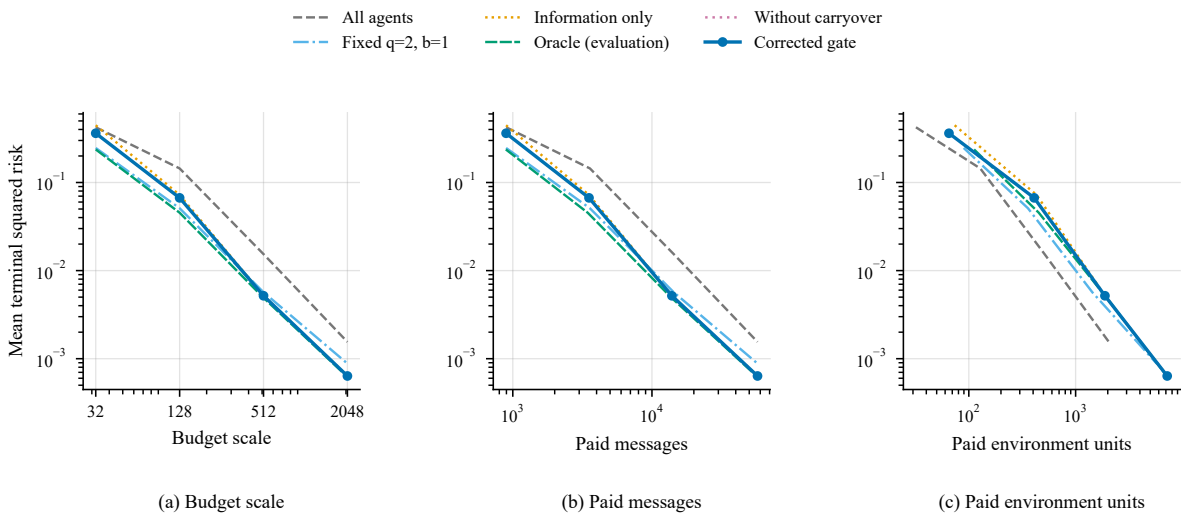
coefficient k_a is common to the hypotheses, as their persistence parameter is the same. Write $\mathcal{C}_j = \mathbb{E}_j[(m_j^2 - s_j^2)\mathbf{1}_{E_j}]$. Symmetry and association give

$$R_0 = J_{0,a_0} + p_0\Delta J_0 + \Delta_k\mathcal{C}_0, \tag{S22}$$

$$R_1 = J_{1,a_1} + p_1\Delta J_1 - \Delta_k\mathcal{C}_1, \tag{S23}$$

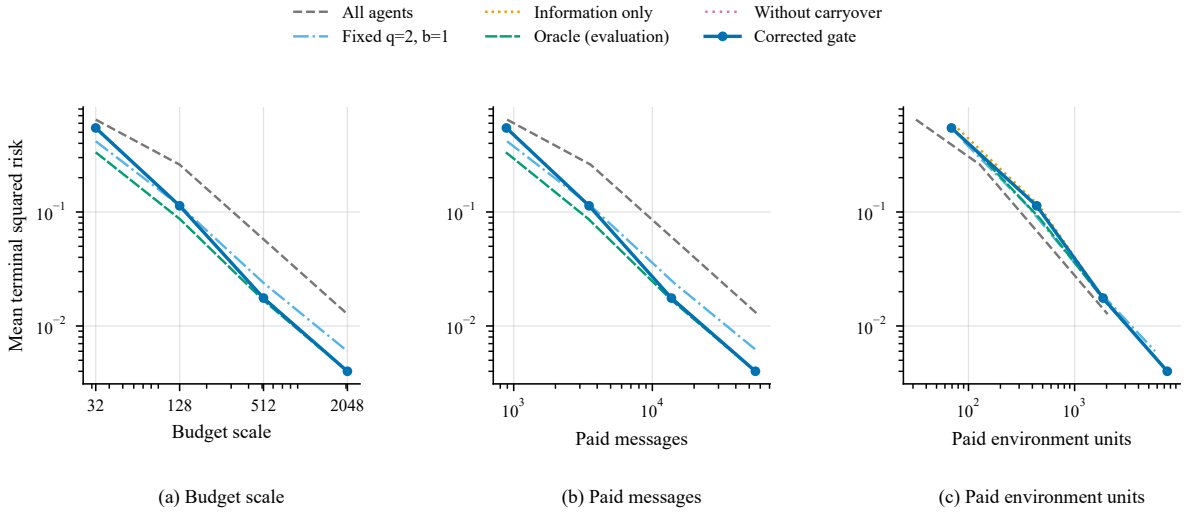
$$\mathcal{C}_0 \geq 0, \quad \mathcal{C}_1 \leq 0. \tag{S24}$$

To see this, subtract Eq. (S3) for the wrong and correct programs. The decision event is even in Y, whereas m_j is linear and odd, so $\mathbb{E}_j[m_j\mathbf{1}_{E_j}] = 0$, including nonzero deterministic initialization. Whiten Y under each true instance and orthogonally diagonalize the positive semidefinite decision quadratic. The resulting coordinates G_i are independent standard normal variables and the outside event is increasing in each G_i^2. Writing $m_j = \sum_i v_i G_i$, coordinate sign symmetry eliminates all cross products. Each diagonal covariance $\mathrm{Cov}(G_i^2, \mathbf{1}_{\mathrm{outside}})$ is nonnegative by association of independent variables [45], Theorem 2.1. For completeness, condition on the other squares and use $2\,\mathrm{Cov}(f(X), g(X)) = \mathbb{E}[(f(X) - f(X'))(g(X) - g(X'))] \geq 0$ for increasing f, g and an independent copy X'. The inside event is the complement, which proves the other sign. Thus $\Delta_k > 0$ permits harmful carryover, $\Delta_k < 0$ permits beneficial carryover, and $\Delta_k = 0$ removes it exactly; strictness needs a nonconstant event



(a) Budget scale (b) Paid messages (c) Paid environment units

Fig. 4. Direct validation, averaged delayed affine learner and low instance; every scheduled update is charged, including withheld terminal updates.



(a) Budget scale (b) Paid messages (c) Paid environment units

Fig. 5. Direct validation, averaged delayed affine learner and high instance; exact overlap of the two gates is retained without graphical offsets.

and an aligned posterior component. In the implemented model $\Sigma_j = I + Q\theta_j R_\lambda$, covariance ordering follows from $\theta_1 > \theta_0$.

B. Classical tail moments and a learner-specific corollary

Let φ and Φ be the standard normal density and distribution function. For $0 < p < 1$, define

$$t_p = \Phi^{-1}(1 - p/2), \qquad h_+(p) = 2t_p\varphi(t_p),$$
$$u_p = \Phi^{-1}((1 + p)/2), \qquad h_-(p) = 2u_p\varphi(u_p).$$

Set both $h_\pm(0) = 0$ by continuity. The truncated-normal moment calculation is classical [46], Sections 2–3 and 5. For any event E of probability p and $m \sim \mathcal{N}(0, s^2)$,

$$-s^2 h_-(p) \le \mathbb{E}[(m^2 - s^2)\mathbf{1}_E] \le s^2 h_+(p). \qquad \text{(S25)}$$

This also allows E to depend on other variables. Indeed, condition its indicator on m to obtain an acceptance function $g(m) \in [0, 1]$ of Gaussian mass p. With $f(m) = m^2 - s^2$, let g_* select the tails $|m| > st_p$ and $c = s^2(t_p^2 - 1)$. Then $(f - c)(g_* - g) \ge 0$ pointwise, and the equal-mass constraint gives $\mathbb{E}[fg_*] \ge \mathbb{E}[fg]$. Selecting the central interval $|m| \le su_p$ proves the minimum in the same way. Integration by parts gives

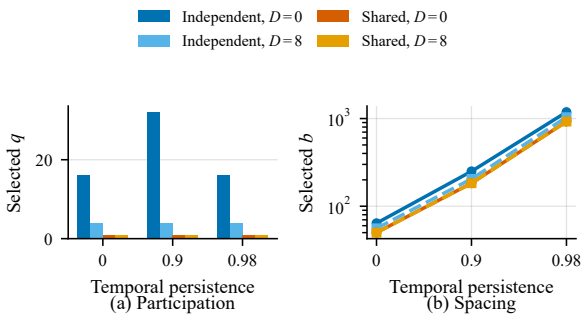


Fig. 6. Joint actions selected by the finite-time Lyapunov certificate. The left panel reports participation and the right panel reports Markov spacing; persistence values occupy categorical positions.

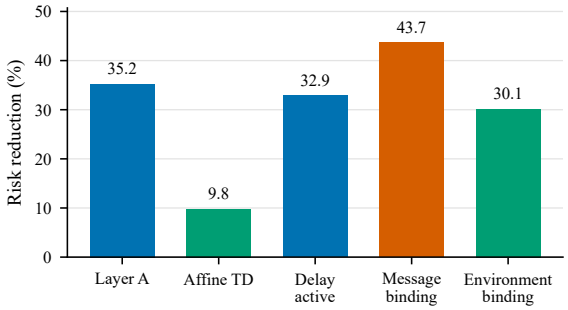


Fig. 7. Relative risk reductions of the learning-aware rule over information-only probing. All percentages use information-only risk as the denominator.

$\int_t^\infty (x^2-1)\varphi(x)\,dx = t\varphi(t)$, establishing Eq. (S25) and its sharp scalar constants.

Proposition 3 (Ordered-Gaussian continuing-risk corollary)**.** *Under the fixed-program, fixed-probe and centered ordered-covariance conditions above, suppose* $p_j \le \delta_j \le 1/4$. *Then*

$$R_0 \le J_{0,a_0} + \delta_0[\Delta J_0]_+ + [\Delta_k]_+ s_0^2 h_+(\delta_0), \qquad \text{(S26)}$$
$$R_1 \le J_{1,a_1} + \delta_1[\Delta J_1]_+ + [\Delta_k]_+ s_1^2 h_-(\delta_1). \qquad \text{(S27)}$$

Proof. On $[0,1/4]$, $h'_+(p) = t_p^2 - 1 \ge 0$ and $h'_-(p) = 1 - u_p^2 \ge 0$. Combine Eq. (S24) with Eq. (S25) and $p_j \Delta J_j \le \delta_j [\Delta J_j]_+$. When $\Delta_k \le 0$, the transfer term is nonpositive under both instances and can be omitted from an upper bound. □

For small δ, $h_+(\delta) = O(\delta \log(1/\delta))$ and $h_-(\delta) = O(\delta)$. The sharpness is for the scalar event-moment problem, not every finite probe geometry. Data-dependent probe stopping, noncentered likelihoods, sign-sensitive classifiers, and data-selected program weights require a different analysis; main Theorem 5 remains available at its stated scope.

C. Why independent-mixture transport can be insufficient

Let G, W be independent standard normals, $Y = \sqrt{3}(\rho G + \sqrt{1-\rho^2}W)$, and choose between the frozen programs $Z_0 = 1$ and $Z_1 = (1+Y)/2$ using $E = \{|G| > t\}$. Both unconditional squared risks equal one, yet

$$\mathbb{E} Z^2_{\text{selected}} = 1 + \frac{3}{4}\rho^2\, 2t\varphi(t) > 1 \qquad \text{(S28)}$$

for $\rho \ne 0$ and $0 < t < \infty$. Conditioning on G gives $\mathbb{E}[Z_1^2 \mid G] = 1 + (\sqrt{3}/2)\rho G + (3/4)\rho^2(G^2-1)$; the odd term cancels and Eq. (S25) evaluates the rest. At $t = 3$ and $\rho = 1/2$, the event probability is 0.002699796 and the risk is 1.004985829. An independent mixture incorrectly gives one even with the exact decision probability, because its unconditional branch-risk gap is zero. This establishes the need to account for selection dependence in this linear Gaussian class, not necessity of a positive penalty when its signed transfer is beneficial. It is an analytical example outside the registered family grid, not a counterexample to both tests of the executed AND-admission rule or a claim about cost-optimal branch choices.

D. Classical decision comparison with paid continuing learners

Fix the public family, learner, step rule and full budget, and let $\mathcal{A}_{\text{fix}}$ contain all original full-budget fixed (q,b) programs. For a stated prior $\pi \in (0,1)$, define

$$I_\pi = \min_{a \in \mathcal{A}_{\text{fix}}} \{(1-\pi)J_0(a) + \pi J_1(a)\},$$
$$O_\pi = (1-\pi)\min_a J_0(a) + \pi \min_a J_1(a),$$
$$V_\pi = I_\pi - O_\pi.$$

This is the finite-action value-of-perfect-information comparison already used in [38], Section 2.6, with loss rather than utility. The finite catalogue gives $V_\pi > 0$ exactly when the two risk functions have no common minimizer: each fixed action's excess over O_π is a positively weighted sum of two nonnegative excess risks, and the minimum is attained. If B_j is a valid actual-risk upper bound for the executed policy, main Eq. (42) proves lower prior-weighted risk than every fixed catalogue action, including every regime-independent randomized mixture by linearity. To expose the cost balance, let $L_j = J^{\text{post}}_{j,a_j} - \min_a J_j(a)$ and $D_j = B_j - J^{\text{post}}_{j,a_j}$. The same sufficient condition is

$$V_\pi > (1-\pi)(L_0 + D_0) + \pi(L_1 + D_1). \qquad \text{(S29)}$$

TABLE V
FULL ORIGINAL-GRID, BALANCED-PRIOR POPULATION CERTIFICATES USING THE EXECUTED GATE'S ORIGINAL BOUND, NOT A NEWLY EXECUTED POLICY.

Budget scale	Running mean	Delayed affine
32	0/16	0/16
128	0/16	0/16
512	6/16	2/16
2048	12/16	9/16

Here L_j can be signed unless risk monotonicity in horizon has separately been proved for the affine transient. Thus decision opportunity must exceed remaining-horizon and decision-event risk, not merely produce an accurate classification. The reference O_π allows a regime-informed fixed commitment, not arbitrary state-adaptive programs; $V_\pi = 0$ does not exclude useful same-stream state selection. The comparison fixes the learner, step rule and catalogue and is prior-specific, rather than a statement about all fixed algorithms, unknown priors or minimax risk.

E. Complete original-design diagnostic

This outcome-known deterministic calculation retains all 128 original designs and introduces no trajectory, seed or confidence interval. The catalogues contain 16 actions at $Q = 8$ and 24 at $Q = 32$, including all original spacings and full-budget affordable horizons. Of 120 feasible probes, 44 have potentially harmful transfer, 32 beneficial transfer and 44 zero transfer; eight infeasible designs remain included. The original and sign-refined admission counts are both 113/128. All 44 harmful designs were already admitted under the original enclosure, and removing the penalty does not rescue the seven feasible rejections. These deterministic decisions explain the unchanged no-carryover ablation without attributing its equality to seed noise.

For $\pi = 1/2$, 72/128 designs have positive V_π, while the original gate's bound certifies separation in 29/128, as shown in Table V. Their relative certified margins range from 0.1314 to 7.2032 percent of $I_{1/2}$; this range is not an average empirical effect. All 99 uncertified balanced cases remain; zero of 128 satisfy the separate sufficient minimax comparison $\max_j B_j < \min_a \max_j J_j(a)$. These population-model diagnostics explain conditional usefulness and do not replace the validation's prespecified regime-stratified endpoints. The original adverse low-budget result, 23/256 adverse cell means, identical corrected/no-carryover trajectories, and fixed $(2, 1)$ pooled risk advantage all remain unchanged.

XVII. Additional Known-Dependence Resource Evidence

A. Correlation-limited participation

The participation study varies $q \in \{1, 2, 4, 8, 16, 32\}$, six cross-agent correlation levels, three delays, three message-equivalent budgets, and thirteen step sizes over 32 paired seeds. At $q = 32$, median effective participation is 30.996 for independent trajectories and 1.111 at correlation 0.9. At the largest evaluated budget and delay eight, the resource-optimal participation moves from $q = 16$ under independence to $q = 1$ under high sharing. Using the endpoint action selected for the opposite correlation regime increases error by factors of approximately 4.29 and 3.13, respectively. Equal charged endpoint budgets isolate the value of dependence-matched participation.

B. Finite-time certificate and joint actions

The joint-action study evaluates twelve $(\text{persistence}, \rho, D)$ scenarios with 32 new paired seeds and 1,152 policy runs. The joint controller searches a stability-screened catalogue of $q \in \{1, 2, 4, 8, 16, 32\}$, charged spacings, and scalar steps. All selected actions satisfy $0 < c_{\text{aff}} < 1$ and the declared budget. The finite-time bound beats the no-update risk in all twelve scenarios (12/12). Every one-sided 99 percent bootstrap upper mean lies below its corresponding bound, with a maximum empirical-upper-bound to theorem-bound ratio of 0.305. The median theorem-bound to empirical-mean ratio is 20.77 and the maximum is 228.98. The selected actions show how the certificate responds to correlation, delay, and persistence despite this slack. Mean participation is 12.67 for independent sources and $q = 1$ in every high-sharing scenario. Delay reduces the selected independent-source participation to $q = 4$, while raising temporal persistence from zero to 0.98 increases the median paid spacing by a factor of 18.49. Fig. 6 retains the complete action plot.

C. Held-out resource-indexed convergence curves

Fig. 8 shows the persistence-0.9, zero-delay slice: broad participation accelerates independent-source learning, whereas low participation avoids the shared-noise floor.

XVIII. Proof of the Conditional Initialization Value

Fix the target-invariant probe Y and the deterministic continuing-program catalogue in main Theorem 6. Under

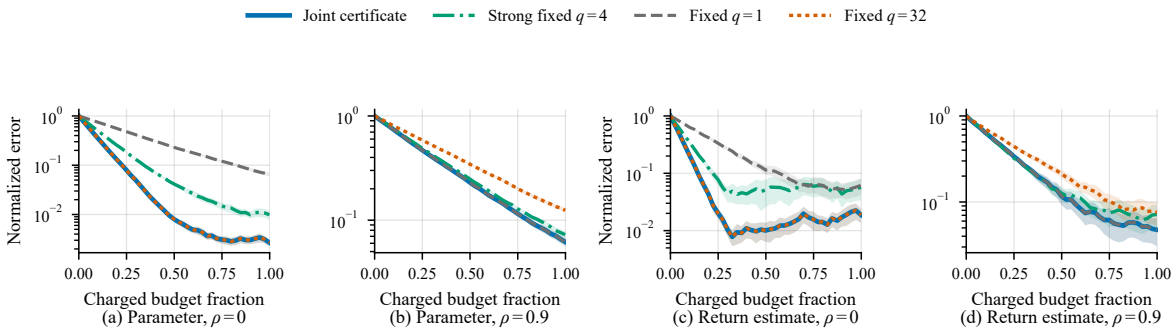


Fig. 8. Held-out resource-indexed convergence for persistence 0.9 and zero delay (64 confirmation seeds; bands are 95 percent confidence intervals). The first two panels show parameter risk and the last two return-estimation risk, compared with the development-selected fixed $q = 4$.

regime j, Gaussian conditioning gives

$$\begin{aligned}\mathbb{E}_{j,e}[Z_a^2 \mid Y = y] = J_{j,a}^r + \zeta_a^2(e^2 - r^2) \\ + 2e\zeta_a k_{j,a} m_j(y) \\ + k_{j,a}^2[m_j(y)^2 - s_j^2].\end{aligned}$$

The observation law and regime posterior do not depend on the unknown fixed e. For any randomized selector, average its risks at $e = r$ and $e = -r$. The odd cross terms cancel, so the maximum risk is at least $\mathbb{E}_g \min_a L_a(Y)$. The finitely many scores L_a are even and measurable, and a fixed first minimizer u is even. Its cross term integrates to zero under each centered Gaussian regime. Its risk is a nondecreasing affine function of e^2 with nonnegative coefficient, and hence its maximum is attained at the endpoints. It attains the lower bound, proving main Eq. (44). This argument uses classical conditional-loss optimization and symmetry, with the continuing learner and initialization-information class stated explicitly. The corresponding joint inference principles appear in [39], [40], [42] and [41].

The remaining proofs separate state-selection value, invariant observability, the cost of paid inference and the fixed-program comparison. The error-contrast identity is classical [43]; the vector reduction uses the minimax theorem of [44].

XIX. A Finite-Budget Ceiling on State-Selection Value

An optimal conditional-risk rule does not by itself imply a practically large improvement. In the scalar Gaussian continuation model one can bound the entire value of using the current noise state, before collecting scientific trajectories.

Fix the same paid probe, remaining budgets, public regime prior and finite catalogue for both rules. Let $J_{j,a}^r$ be the unconditional commitment risk at initialization radius r and let $s_j^2 = \mathrm{Var}_j(m_j(Y))$. Allow a separate common-state transfer coefficient $k_{j,a}$ under each covariance model, including different persistence parameters. The even conditional risk score is

$$\begin{aligned}L_a(y) &= F_a(y) + \sum_j p_j(y) k_{j,a}^2 S_j(y), \\ F_a(y) &= \sum_j p_j(y) J_{j,a}^r, \\ S_j(y) &= m_j(y)^2 - s_j^2.\end{aligned}$$

Let $c(y)$ minimize $F_a(y)$ and let $u(y)$ minimize $L_a(y)$, using fixed measurable tie rules. Both are even selectors and therefore have the initialization-minimax interpretation already proved. In this section $R(v)$ means the maximum prior-weighted risk over the public scalar initialization interval; evenness attains that maximum at $e = \pm r$. Put $\mathcal{R}_j = \max_a k_{j,a}^2 - \min_a k_{j,a}^2$.

$$0 \le R(c) - R(u) \le 4\varphi(1) \sum_j \pi_j \mathcal{R}_j s_j^2. \tag{S30}$$

The result is a finite-model upper bound on state information beyond the model posterior; it does not bound the value of identifying the covariance regime itself. It remains valid for an observable mean-invariant Gaussian probe, using that probe's own explained variances.

For the proof, pointwise optimality gives

$$\begin{aligned}0 \le L_c - L_u &\le \sum_j p_j(k_{j,c}^2 - k_{j,u}^2) S_j \\ &\le \sum_j p_j \mathcal{R}_j |S_j|.\end{aligned}$$

The first inequality uses $L_u \le L_c$, and the second uses $F_c \le F_u$. Since $p_j(y)g(y) = \pi_j f_j(y)$,

$$\mathbb{E}_g[p_j(Y)|S_j(Y)|] = \pi_j \mathbb{E}_j|m_j(Y)^2 - s_j^2| = 4\varphi(1)\pi_j s_j^2.$$

The last equality follows from the centered Gaussian identity $\mathbb{E}|G^2 - 1| = 4\varphi(1)$; it also holds at $s_j = 0$ by

continuity. Integrating proves Eq. (S30). The stop action has $k = 0$ and fits the same bound.

For a running mean with N applied observations and model-specific sample persistence $\rho = \lambda_j^b$, $|\rho| < 1$,

$$|k_{j,a}| = \frac{|\rho(1-\rho^N)|}{(N+1)|1-\rho|} \le \frac{|\rho|}{(N+1)(1-|\rho|)}.$$

For an averaged stable affine-delay learner, assume its impulse response is absolutely summable with sum of absolute values at most G_*. The exact averaging weights obey $|w_t| \le \eta G_*/N$, giving

$$|k_{j,a}| \le \frac{\eta G_* |\rho|}{N(1-|\rho|)}.$$

Absolute summability is required here; it is not inferred merely from a numerical finite-horizon check. For the registered affine steps it follows from the constant-delay characteristic polynomial $z^{D+1} - z^D + \alpha$, with $\alpha = \eta\mu = 0.1/(D+1)$. All roots are inside the unit disk for $0 < \alpha < 2\sin[\pi/(4D+2)]$. Indeed for positive small α the root initially at one moves inside while the other roots remain near zero. At a unit-circle crossing $z = e^{i\omega}$ the equation $z^D(z-1) = -\alpha$ gives a smallest positive crossing magnitude $2\sin[\pi/(4D+2)]$; root continuity excludes an earlier exit. The registered α is strictly below that magnitude, since $2\sin[\pi/(4D+2)] \ge 2/(2D+1) > 0.1/(D+1)$. The impulse generating function $[1 - x + \alpha x^{D+1}]^{-1}$ is therefore analytic on a disk of radius greater than one, proving absolute summability. The finite registered delay set has a finite uniform G_*. If non-stop catalogue horizons are at least a positive constant times the budget scale B, and persistence and impulse bounds are uniform over that catalogue, the state-selection value is $O(B^{-2})$. Relative to a best fixed risk of order B^{-1} this contribution is at most $O(B^{-1})$. For these stable averaged learners, with finite participation, nonzero private variance, persistence bounded below one and affordable horizons proportional to B, that fixed-risk order follows directly: $\sum_t w_t$ tends to a nonzero constant, Cauchy–Schwarz gives a private-variance lower bound proportional to N^{-1}, and the weight bound and summable temporal covariance give the matching upper bound. Covariance-regime adaptation can change the leading B^{-1} variance coefficient, so these two information roles must not be conflated. The ceiling predicts small state-driven policy differences at long horizons while leaving covariance-regime adaptation as a potential leading-order effect.

A. The second-order state ceiling is attained

The order B^{-2} can be attained by resource programs of the same running-mean learner. Take a single Gaussian regime with $\theta = 1/3$, $\lambda = 1/5$, $Q = 8$, $h = 8$, $D = 0$ and $r^2 = 49/50$. A two-observation all-actor probe uses 32 messages and two clock units. With total budgets $(80B+32, 32B+2)$, its two continuing programs $(q,b) = (2,1)$ and $(8,1)$ have applied horizons $8B$ and $5B$. Both have leading risk $1/(8B)$. The invariant contrast is one dimensional, with explained terminal-state variance

$$s^2 = \frac{[\theta(1-\lambda)]^2}{2[\theta(1-\lambda)+1/8]} = \frac{64}{705}.$$

Let c select the smaller unconditional commitment risk and u the even conditional-risk minimum over these two programs. For all sufficiently large integer B, c selects $(2,1)$. The exact geometric risk formula yields

$$B^2(J^r_{8,1} - J^r_{2,1}) \longrightarrow a = \frac{19}{320000},$$

whereas $B^2(k^2_{8,1} - k^2_{2,1})s^2 \longrightarrow b = 13/94000$. Here $J^r_{q,b}$ and $k_{q,b}$ name commitment risk and state transfer, not a new model index. Since $0 < a < b$, the conditional scores cross on a set of positive Gaussian probability. Writing $G \sim \mathcal{N}(0,1)$, pointwise minimization and dominated convergence give

$$\lim_{B\to\infty} B^2[R(c) - R(u)] = b\,\mathbb{E}\left[\left(\frac{1187}{2080} - G^2\right)_+\right] > 0.$$

For domination, the scaled risk gaps and state coefficients are bounded by the geometric expansions, so the positive part is bounded by a constant times $1 + G^2$. Thus the state-value ceiling has the sharp budget order in this program class, even with no unknown regime to identify. The construction establishes strict second-order value, not a first-order gain or confirmation of the separate empirical state-selection test.

XX. Separating Identification Value, State Value, and the Paid Horizon

The exact initialization-minimax envelope can be compared with the complete full-budget fixed catalogue without confusing nominal regime value and actual risk. For one fixed probe, let I_{full} be the best full-budget fixed risk, $I_{\text{post}} = \min_a \sum_j \pi_j J^r_{j,a}$ the best observation-independent program after paying the probe, $R_{\text{nom}} = \mathbb{E}_g \min_a F_a(Y)$, and $R_{\text{rob}} = \mathbb{E}_g \min_a L_a(Y)$. Let $c(Y)$ minimize $F_a(Y)$ and $u(Y)$ minimize $L_a(Y)$, with fixed even tie rules. The actual risk of c is $R_c = \mathbb{E}_g L_{c(Y)}(Y)$, which need not equal R_{nom}. All post-probe quantities use exactly the same remaining budgets and catalogue, including stop. Define

$$\begin{aligned} C &= I_{\text{post}} - I_{\text{full}}, \\ V_{\text{reg}} &= I_{\text{post}} - R_{\text{nom}}, \\ V_{\text{state}} &= R_c - R_{\text{rob}}, \\ T &= \mathbb{E}_g \sum_j p_j(Y) k^2_{j,c(Y)} S_j(Y). \end{aligned}$$

Then

$$I_{\text{full}} - R_{\text{rob}} = V_{\text{reg}} + V_{\text{state}} - C - T. \tag{S31}$$

Here $V_{\text{reg}}, V_{\text{state}} \geq 0$ by pointwise minimization. The signed term T is essential: regime-posterior decisions and continuing state risk use the same probe. Treating the nominal posterior score as actual risk would omit precisely the dependence at issue. The quantity C is the exact, possibly signed, change in the best fixed risk caused by the paid remaining horizon; no monotonicity of the affine learner in its horizon is assumed. This decomposition is an accounting identity using classical decision value, not a new general information-value theorem.

Let K be the finite state-value ceiling proved above, and define the post-probe perfect-regime fixed risk $I_{\text{oracle}} = \sum_j \pi_j \min_a J^r_{j,a}$. Because a posterior-weighted average is at least the posterior average of the individual regime minima, $R_{\text{nom}} \geq I_{\text{oracle}}$. For each j, $\mathbb{E}_g[p_j S_j] = 0$. Centering $k^2_{j,c}$ at the midpoint of its model-specific catalogue range and summing gives $|T| \leq K/2$. Pointwise $\min_a L_a \geq \min_a F_a + \sum_j p_j \min_a(k^2_{j,a} S_j)$. The expectation of the latter sum is $-\frac{1}{2}\sum_j (\max_a k^2_{j,a} - \min_a k^2_{j,a})\mathbb{E}_g[p_j|S_j|] \geq -K/2$. Consequently

$$\begin{aligned} 0 \leq V_{\text{state}} &\leq K, \\ 0 \leq V_{\text{reg}} &\leq I_{\text{post}} - I_{\text{oracle}}, \\ |T| &\leq K/2, \\ I_{\text{full}} - R_{\text{rob}} &\leq I_{\text{full}} - I_{\text{oracle}} + K/2. \end{aligned}$$

Thus $I_{\text{oracle}} - K/2 \geq I_{\text{full}}$ rules out any strict improvement by any randomized measurable selector following this fixed probe, even if its regime inference is optimal. For a desired relative gain $\epsilon > 0$, the stronger necessary condition is $I_{\text{full}} - I_{\text{oracle}} + K/2 \geq \epsilon I_{\text{full}}$. Conversely $V_{\text{reg}} > C + K/2$ suffices for a strict paid improvement, since the state-value contribution is nonnegative and $T \leq K/2$. Neither condition ranges over different probes, estimators, step schedules or unknown covariance models. These pre-data criteria explain both when identification can matter and when more validation seeds cannot rescue a structurally uneconomic probe.

XXI. Observable Calibration with an Unknown Target

The centered-noise pilot assumption can be replaced by an observable invariant statistic in a Gaussian location model. The projection and generalized least-squares identities below are classical linear-model tools; the issue is whether their statistic supplies the continuing learner's posterior state without revealing the unknown target.

A. Eliminating the target without an oracle

Let the paid probe be $Z = X\vartheta + W_j$, where the unknown target ϑ is fixed, the public matrix $X \in \mathbb{R}^{n\times p}$ has full column rank and $n > p$, and $W_j \sim \mathcal{N}(0, \Sigma_j)$ under a public finite collection of covariance models. Choose a deterministic full-row-rank $D \in \mathbb{R}^{(n-p)\times n}$ satisfying $DX = 0$. Then $Y = DZ = DW_j$ is observable and its law is independent of ϑ. Its covariance $K_j = D\Sigma_j D^\top$ is positive definite. If the continuing common state C has cross-covariance $B_j = \mathrm{Cov}_j(C, W_j)$ and variance T_j, Gaussian conditioning gives

$$\begin{aligned} \mathbb{E}_j[C \mid DZ] &= B_j D^\top K_j^{-1} DZ, \\ \mathrm{Var}_j(C \mid DZ) &= T_j - B_j D^\top K_j^{-1} D B_j^\top. \end{aligned}$$

The mean-invariant risk selector therefore has the previous conditional-loss and initialization-minimax guarantees when its input is DZ and its model posterior uses K_j. No subtraction of the true target and no improper target prior are needed. The guarantee is scoped to selectors using this invariant statistic and the specified frozen continuing-program catalogue; it is not optimality over estimators using every part of the raw probe.

B. Projection and determinant identities

Define

$$\Pi_j = \Sigma_j^{-1} - \Sigma_j^{-1} X (X^\top \Sigma_j^{-1} X)^{-1} X^\top \Sigma_j^{-1}.$$

Then $\Pi_j = D^\top (D\Sigma_j D^\top)^{-1} D$ and

$$\det(D\Sigma_j D^\top) = \det \Sigma_j \det(X^\top \Sigma_j^{-1} X) \frac{\det(DD^\top)}{\det(X^\top X)}.$$

For the first identity, whiten Σ_j; both sides become the orthogonal projection onto the complement of the column space of $\Sigma_j^{-1/2} X$, conjugated by $\Sigma_j^{-1/2}$. For the determinant identity, take orthonormal bases for the columns of X and the rows of D. In that orthogonal basis let C be the covariance block on the contrast coordinates. The block inverse and determinant formulas give $\det C = \det \Sigma_j \det(U_X^\top \Sigma_j^{-1} U_X)$, where $U_X = X(X^\top X)^{-1/2}$. Restoring the contrast basis multiplies by $\det(DD^\top)$ and gives the stated formula. Thus $Z^\top \Pi_j Z$, $B_j \Pi_j Z$ and $B_j \Pi_j B_j^\top$ are exactly target invariant.

For a scalar unknown level, $X = \mathbf{1}$ and D can be the first-difference matrix. Its two Gram determinants both equal n, so the ratio cancels. Writing $S_j = \mathbf{1}^\top \Sigma_j^{-1} \mathbf{1}$, one obtains

$$\log\det K_j = \log\det \Sigma_j + \log S_j.$$

This reduces an $(n-1)$-dimensional determinant to the ordinary innovations determinant and a scalar generalized least-squares information.

C. Linear-time scalar Markov implementation

For the scalar common-state Markov model with known λ, noise variance θ_j and private observation variance $1/Q$, run the usual innovations filter twice with the same deterministic variance sequence: once on the raw probe and once on the all-one vector. Let their whitened innovations be u_t and v_t, with innovation variance F_t, and let the two final filtered common-state means be m_Z and m_1. Accumulate

$$S = \sum_t v_t^2/F_t, \quad T = \sum_t u_t v_t/F_t, \quad U = \sum_t u_t^2/F_t.$$

Then the invariant quadratic form is $U - T^2/S$, the invariant common-state mean is $m_Z - m_1 T/S$, and its explained variance is $\theta_j - P_j - m_1^2/S$. The formulas follow by substituting the innovations factorization into Π_j and using the filtered-state identity $\mathrm{Cov}(C, Z)\Sigma_j^{-1} Z = m_Z$. Centering the raw probe by its first entry before the calculations leaves DZ unchanged and can reduce cancellation for large unknown target levels. For the first-difference coordinates, the exact log density is

$$-\tfrac{1}{2}\{(n-1)\log(2\pi) + \log\det\Sigma_j + \log S + U - T^2/S\}.$$

The posterior-state and likelihood calculations require linear time and constant filter-state storage per covariance hypothesis. All probe observations still incur their message and environment costs, even though the level component is discarded by the selector.

The exact Gaussian affinity for two invariant probe laws is

$$\rho_D = \frac{\det K_0^{1/4} \det K_1^{1/4}}{\det[(K_0 + K_1)/2]^{1/2}}.$$

For $\Sigma_j = \theta_j R_\lambda + I/Q$, the covariance midpoint has the same form, so three variance/ones-filter recursions evaluate this affinity. A zero-threshold likelihood rule has each directional error probability at most ρ_D, by applying Markov's inequality to the square root likelihood ratio under each law. The old full-noise probe's affinity cannot be reused as if projecting out the target were free information.

D. A finite bound on the information cost of removing the target

Suppose $a_j I \preceq \Sigma_j \preceq b_j I$ uniformly in the probe length, with $a_j > 0$, and let b_{mid} bound the covariance midpoint from above. Write ρ for the full centered-noise affinity and ρ_D for the contrast affinity. Then

$$1 \le \frac{\rho_D}{\rho} \le C_p := \frac{b_{\mathrm{mid}}^{p/2}}{(a_0 a_1)^{p/4}}. \tag{S32}$$

For the lower bound, complete the contrast coordinates to an invertible linear transformation and apply Cauchy–Schwarz along each discarded coordinate fiber; marginalizing those coordinates increases Gaussian affinity. For the upper bound, set $U = X(X^\top X)^{-1/2}$. The determinant identity gives

$$\frac{\rho_D}{\rho} = \frac{\det(U^\top \Sigma_0^{-1} U)^{1/4} \det(U^\top \Sigma_1^{-1} U)^{1/4}}{\det(U^\top \Sigma_{\mathrm{mid}}^{-1} U)^{1/2}}.$$

The eigenvalues of $U^\top \Sigma_j^{-1} U$ lie between $1/b_j$ and $1/a_j$; substitution proves Eq. (S32). For a fixed nuisance dimension p, the bound is independent of the probe length. Therefore removing an unknown target preserves any existing exponential affinity rate: if $\rho(n) \le K e^{-In}$, then $\rho_D(n) \le C_p K e^{-In}$. The sufficient length calculated from this exponential bound incurs an additive term of order $\log C_p / I$, subject also to $n > p$ and full column rank. This statement does not bound the difference between the two exact minimal confidence-probe lengths without further finite-length control. This is a determinant/spectral consequence of classical Gaussian error contrasts, not a new general invariance or data-processing principle.

For the registered scalar Markov covariance $\Sigma_j = \theta_j R_\lambda + I/Q$, valid uniform choices are

$$a_j = Q^{-1} + \theta_j \frac{1-\lambda}{1+\lambda}, \quad b_j = Q^{-1} + \theta_j \frac{1+\lambda}{1-\lambda}.$$

They follow by bounding the quadratic form of the stationary AR(1) covariance using its spectral density. The constant can grow with nuisance dimension or near-unit persistence; it does not justify cost-free calibration in those regimes. The information-rate bound also does not preserve the explained common-state variance, which must be recomputed from $B_j \Pi_j B_j^\top$.

E. Why more data are not automatically justified

At the same probe schedule, costs and continuation catalogue, observing the full centered noise W is more informative than observing DW. Any invariant selector can be implemented from W by computing DW. Consequently the best full-noise initialization-minimax risk is no larger than the best invariant risk. This follows from inclusion of policy classes and does not require a new empirical experiment or an unrestricted estimation claim. The centered-pilot and unknown-target invariant protocols therefore evaluate distinct information classes. If a new probe schedule is chosen, its different remaining horizons must be recomputed; no unproved monotonicity of the affine learner in its horizon is used.

XXII. First-Order Regime Value Versus Second-Order State Value

The state-value ceiling does not rule out a substantial benefit from identifying the covariance regime: that information can change the leading variance coefficient

of a resource program. The distinction can be made exact against every affordable fixed training prefix, rather than only against programs that spend the whole budget.

Consider the running mean with a fixed public initial-error radius $r > 0$, finite positive participation and spacing sets, bounded delay D, and $0 \le \lambda \le 1-\gamma < 1$. Write the message and clock budgets as $\lfloor \beta_m B \rfloor$ and $\lfloor \beta_e B \rfloor$, where $\beta_m, \beta_e > 0$. For program $a = (q, b)$ define

$$\begin{aligned} a_a &= \min\{\beta_m/(h+q), \beta_e/b\}, \\ v_j(a) &= q^{-1} + \theta_j \frac{1+\lambda^b}{1-\lambda^b}, \\ K_j(a) &= v_j(a)/a_a. \end{aligned}$$

All programs retain the same running-mean rule; a fixed alternative may choose any deterministic scheduled prefix, including zero updates, within its budgets. Let

$$\begin{aligned} K_\pi &= \min_a \sum_j \pi_j K_j(a), \\ a_j^* &\in \arg\min_a K_j(a), \\ \Delta &= K_\pi - \sum_j \pi_j K_j(a_j^*). \end{aligned}$$

For two regimes and a fixed tie rule put $A = \sum_j \pi_j [K_j(a_{1-j}^*) - K_j(a_j^*)]$. This nonnegative quantity bounds the leading cost of a wrong commitment. The model laws, prior, initialization radius and probe schedule are public; the realized regime and initial error are not.

Theorem S1 (Paid separation from every fixed training prefix)**.** *Suppose a deterministic, target-invariant Gaussian probe has length n, each directional likelihood error is at most $0 < \delta < 1/2$, and the continuing programs commit to a_j^* on its decision. If $\Delta > \delta A$, its signed same-stream upper certificate is below the minimum worst-initialization prior risk of every full-budget fixed program and every observation-independent randomized mixture for all sufficiently large B. More precisely the certified advantage is at least*

$$\frac{\Delta - \delta A}{B} - \frac{C(n+1)}{B^2},$$

for a finite model/catalogue-dependent C, once all post-probe horizons are positive fractions of B. If instead the regimes share a leading-coefficient minimizing action, then $\Delta = 0$, and no selector from the maximal-affordable-horizon post-probe catalogue plus stop, following any such probe with $n = O(\log B)$, can improve on the best fixed prefix by more than $O((n+1)/B^2)$.

The positive claim is relative to a fixed learner and a complete resource-and- prefix class, not to arbitrary estimators or learning-rate sequences. The negative claim is an order statement for a specified sublinear paid probe; it is not an impossibility theorem over all sequential sensing policies.

A. Proof and the role of every hypothesis

For a fixed action and N applied observations, the exact running-mean risk is

$$R_{j,a}(N) = \frac{r^2 + N/q + \theta_j[N + 2\sum_{s=1}^{N-1}(N-s)\lambda^{bs}]}{(N+1)^2}.$$

The geometric-sum formula gives $R_{j,a}(N) = v_j(a)/N + O(N^{-2})$, uniformly on the finite catalogue. It also gives $R_{j,a}(N+1) - R_{j,a}(N) = -v_j(a)/N^2 + O(N^{-3})$ with an exponentially small remainder. Since $v_j(a) > 0$, each action's risk, and each prior average of it, decreases strictly beyond a fixed horizon. Every finite early prefix has positive risk, whereas its maximal affordable prefix has risk tending to zero. Thus the optimum over all prefixes eventually uses each candidate action's maximal horizon, and its global optimum is $I_{\text{full}} = K_\pi/B + O(B^{-2})$. The same endpoint initialization maximizes all scalar fixed risks; randomizing fixed programs cannot lower their minimum prior risk.

Subtracting a length-n paid probe changes each maximal horizon from $a_a B + O(1)$ to $a_a B + O(n+1)$, including rounding and bounded withheld updates. The correct-program prior risk is therefore $\sum_j \pi_j K_j(a_j^*)/B + O((n+1)/B^2)$. The directional-error term is at most $\delta A/B + O((n+1)/B^2)$. The even invariant likelihood decision cancels the unknown initialization's odd term; the signed continuing-state correction is $O(B^{-2})$, because $k_a = O(B^{-1})$ and the probe's explained variance is at most θ_j. The constants in its truncated-normal factors are finite at fixed δ. Combining these facts with the fixed-prefix expansion proves the lower bound on the certified advantage.

For the negative claim, a common leading minimizer gives $K_\pi = \sum_j \pi_j \min_a K_j(a)$. The post-probe perfect-regime fixed risk consequently has that same leading coefficient and a remainder $O((n+1)/B^2)$. The preceding oracle-plus-half-state-ceiling bound applies to every measurable randomized selector, whose state-value ceiling is $O(B^{-2})$. Subtracting the fixed-prefix risk proves the stated upper order. No monotonicity of an averaged affine learner is invoked in this running-mean theorem.

B. Existence of an observable finite-confidence probe

For $\Sigma_j(n) = \theta_j R_\lambda(n) + I/Q$, with distinct positive variances, the eigenvalues of $R_\lambda(n)$ lie in a fixed positive compact interval. Both covariance matrices are functions of the same $R_\lambda(n)$. Each eigen-coordinate contributes a Gaussian affinity factor strictly below one, uniformly on that interval. Thus the full centered-noise affinity is at

most κ^n for some $\kappa < 1$. The contrast determinant bound gives $\rho_D(n) \le C_1\kappa^n$ for the unknown scalar target. Any fixed confidence $\delta > 0$ therefore has a finite, observable paid probe. Its actual directional bound is recomputed after projection, not inherited from the centered-noise experiment.

C. An explicit sufficient budget threshold

The sufficiently-large-budget statement has computable, albeit conservative, constants. All action maxima in this subsection range over non-stop resource pairs; the zero-update program is treated separately. Write $\rho_a = \lambda^{b_a}$ and define

$$c_{j,a} = r^2 + 2v_j(a) + \frac{2\theta_j\rho_a}{(1-\rho_a)^2}.$$

The geometric sum gives the exact identity

$$R_{j,a}(N) = \frac{r^2 + Nv_j(a) - 2\theta_j\rho_a(1-\rho_a^N)/(1-\rho_a)^2}{(N+1)^2},$$

and hence $|R_{j,a}(N) - v_j(a)/N| \le c_{j,a}/N^2$ for every $N \ge 1$. Let p_m, p_e be the probe's full paid message and clock charges, including any fixed startup cost, and put

$$L_a(p) = D+1+\max\{(1+p_m)/(h+q_a), (1+p_e)/b_a\}.$$

For the declared FIFO convention, rounding the two budgets and subtracting the D withheld updates gives $a_aB - L_a(p) \le N_a^{\mathrm{post}} \le a_aB$. Thus $B \ge 2L_a(p)/a_a$ implies

$$\left| J_{j,a}^r - \frac{K_j(a)}{B} \right| \le \frac{C_{j,a}(p)}{B^2},$$
$$C_{j,a}(p) = \frac{4c_{j,a} + 2v_j(a)L_a(p)}{a_a^2}.$$

The all-prefix lower bound can also avoid an unspecified eventual-monotonicity threshold. For each action use the prior averages $\bar v_a = \sum_j \pi_j v_j(a)$ and $\bar c_a = \sum_j \pi_j c_{j,a}$, and set $n_{0,a} = \lceil 2\bar c_a/\bar v_a \rceil$. For $N \ge n_{0,a}$ the expansion lower bound is at least $\bar v_a/(2N)$. If $N < N_{\max,a}/4$ this exceeds $\bar v_a/N_{\max,a}$; if $N \ge N_{\max,a}/4$, the expansion instead bounds the remainder by $16\bar c_a/N_{\max,a}^2$. For $N < n_{0,a}$, including stop, the initialization term is at least $r^2/(n_{0,a}+1)^2$. Consequently every prefix obeys

$$\bar R_a(N) \ge \frac{\bar v_a}{N_{\max,a}} - \frac{16\bar c_a}{N_{\max,a}^2}$$

whenever $N_{\max,a} \ge \bar v_a(n_{0,a}+1)^2/r^2$. It follows that

$$I_{\mathrm{full}} \ge \frac{K_\pi}{B} - \frac{C_F}{B^2}, \quad C_F = \max_a \frac{64\bar c_a}{a_a^2},$$

provided $B \ge B_F$, where

$$B_F = \max\left\{ \frac{K_\pi}{r^2}, \max_a \frac{2L_a(0)}{a_a}, \max_a \frac{2\bar v_a(n_{0,a}+1)^2}{a_a r^2} \right\}.$$

The zero-update program has risk r^2 and is covered by the first term.

Let $c_j = a_j^*$, $w_j = a_{1-j}^*$, and $\kappa = \max_a 4\rho_a^2/[a_a^2(1-\rho_a)^2]$. At $B \ge \max_a 2L_a(p)/a_a$, each common-state coefficient has $k_a^2 \le \kappa/B^2$. The explained variance is at most θ_j, and the even decision has centered-square selection correction at most $4\varphi(1)\theta_j\kappa/B^2$. This also bounds the corresponding signed truncated-normal correction from above. Using the directional error bound δ and $K_j(w_j) \ge K_j(c_j)$ therefore gives a certified upper risk

$$U \le \frac{\sum_j \pi_j K_j(c_j) + \delta A}{B} + \frac{C_U}{B^2},$$

where

$$C_U = \sum_j \pi_j \{C_{j,c_j}(p) + \delta[C_{j,c_j}(p) + C_{j,w_j}(p)] + 4\varphi(1)\theta_j\kappa\}.$$

Thus a completely explicit sufficient threshold is

$$B \ge \max\{B_F, \max_a 2L_a(p)/a_a\}, \quad B > \frac{C_F + C_U}{\Delta - \delta A},$$

when $\Delta > \delta A$. All constants use public model, catalogue and paid-probe quantities. They may be much more conservative than evaluating the exact finite-budget certificate; no small practical threshold or observed improvement is inferred from this construction. For a length-n probe with bounded per-observation charges, $C_U = O(n+1)$, recovering the order in the theorem.

D. A rational resource witness on the existing public grid

Take $Q = 8$, $h = 8$, $\theta_0 = 1/20$, $\theta_1 = 2$, $\lambda = 1/5$, balanced prior, $\beta_m = 16$, $\beta_e = 4$, and the original $q \in \{1,2,4,8\}$, $b \in \{1,2,4,8\}$ catalogue. Exact enumeration of its sixteen leading coefficients yields

$$a_0^* = (8,4), \quad K_0(a_0^*) = 1093/6240,$$
$$a_1^* = (2,2), \quad K_1(a_1^*) = 5/3,$$

while the best balanced fixed coefficient is $K_\pi = 773/768$ at $(2,2)$. The wrong-commitment coefficient is $A = 407/1280$. At directional bound $\delta = 1/40$, the sufficient leading margin $\Delta - \delta A$ is strictly positive (approximately 0.07765). Its ratio to K_π is approximately 7.71 percent. This is a model-population, sufficiently-large-budget witness, not a finite- grid empirical gain or a claimed threshold value. All finite budget thresholds and admitted finite-grid designs must still be computed from the actual paid projected probe and exact learner risk.

XXIII. A Finite-Budget Separation with Covariance Uncertainty

This section establishes the probe certificate and the short-cycle refinement used alongside main Theorem 7. The detector and commitments remain fixed throughout the parameter box. The comparison permits every affordable prefix of every resource cycle of period at most three, stop, and observation-independent random mixtures. There are $16 + 16^2 + 16^3 = 4368$ ordered cycle words over the sixteen participation–spacing pairs.

A. A target-invariant probe with uniform error control

For the reference covariance $\Sigma_j = \theta_j R_{1/5} + I/8$, the stationary correlation matrix has eigenvalues in $[2/3, 3/2]$. The covariance ratio in every direction is consequently at least

$$\frac{2(2/3) + 1/8}{(1/20)(2/3) + 1/8} = \frac{175}{19}.$$

Congruence by the full-row-rank first-difference matrix preserves this ratio bound for the seventeen-dimensional contrast covariances. If $\ell \geq 175/19$ is a generalized covariance eigenvalue, its Gaussian affinity factor is $[2\sqrt{\ell}/(1+\ell)]^{1/2} \leq 4/5$. Indeed the factor decreases for $\ell > 1$, and its squared comparison reduces to the rational inequality $2500\ell \leq 256(1+\ell)^2$ at $\ell = 175/19$. The reference affinity is therefore at most $(4/5)^{17} < 1/40$. The square-root likelihood-ratio bound gives each reference directional error at most $1/40$.

For a stationary AR(1) correlation matrix the spectral density is proportional to $(1-\lambda^2)/(1-2\lambda\cos\omega+\lambda^2)$. Its ratio to the reference density has extrema at $\omega = 0, \pi$. Writing $s(\lambda) = (1+\lambda)/(1-\lambda)$, every allowed true common-state covariance relative to the reference is bounded below by

$$\frac{99}{100}\min\left\{\frac{s(39/200)}{s(1/5)}, \frac{s(1/5)}{s(41/200)}\right\} = \frac{47223}{48200} > 1 - \frac{1}{20}$$

and above by $(101/100)s(41/200)/s(1/5) = 24341/23850 < 1 + 1/20$. The Toeplitz quadratic-form representation transfers these spectral bounds to every finite covariance matrix. Adding the unchanged private noise and projecting by the contrast matrix preserve both Loewner bounds. Thus each true contrast covariance satisfies $(1-\varepsilon)K_j \preceq \widetilde{K}_j \preceq (1+\varepsilon)K_j$, with $\varepsilon = 1/20$. The two true persistence parameters need not coincide.

For centered Gaussian laws of dimension d, these covariance bounds imply

$$D_{\mathrm{KL}}(\widetilde{P}_j \| P_j) \leq \frac{d\varepsilon^2}{4(1-\varepsilon)}.$$

To see this, each whitened eigenvalue $t \in [1-\varepsilon, 1+\varepsilon]$ contributes $(t - 1 - \log t)/2$. The largest contribution is at $1-\varepsilon$, and $-\varepsilon - \log(1-\varepsilon) \leq \varepsilon^2/[2(1-\varepsilon)]$ by its power series. The classical total-variation inequality $\|P-Q\|_{\mathrm{TV}}^2 \leq D_{\mathrm{KL}}(P\|Q)/2$ follows by applying the log-sum inequality to an event and its complement, then using the binary divergence's second derivative $1/[t(1-t)] \geq 4$. For $d = 17$, the total variation is at most $\varepsilon\sqrt{17/[8(1-\varepsilon)]} \leq 3/40$. Each true directional error of the unchanged reference detector is therefore at most $1/40 + 3/40 = 1/10$. This is a uniform robustness certificate for specified covariance bands, not an estimator of those bands from the eighteen observations.

B. A sharper lower bound over every fixed prefix

For $\rho = \lambda^b \geq 0$ and $N \geq 1$, put $v = q^{-1} + \theta(1+\rho)/(1-\rho)$ and $t = 2\theta\rho/(1-\rho)^2$. The exact stationary running-mean risk at $|e| = r$ obeys

$$\begin{aligned} R(N) &= \frac{r^2 + Nv - t + t\rho^N}{(N+1)^2} \\ &\geq f(N) := \frac{r^2 + Nv - t}{(N+1)^2}. \end{aligned}$$

For real $x \geq 1$, the derivative of f has numerator $v - 2(r^2 - t) - vx$. It can change sign from positive to negative, but never from negative to positive. Consequently its minimum on $[1, N_{\max}]$ is attained at an endpoint. Including stop yields the all-prefix lower bound

$$\min_{0 \leq N \leq N_{\max}} R(N) \geq \min\{r^2, f(1), f(N_{\max})\}.$$

The bound also holds for a prior average by using the prior averages of v and t. Unlike the earlier asymptotic constants, it requires no eventual-monotonicity threshold. For the parameter box, the exact running-mean risk increases with θ and nonnegative λ because every covariance entry and averaging weight is nonnegative. Thus the same lower bound evaluated at $\theta_{j,-}$ and $\lambda_- = 39/200$ bounds every allowed fixed risk below. At each prior endpoint, minimize it over all sixteen programs using $N_{\max} = \min\{\lfloor 16B/(8+q)\rfloor, \lfloor 4B/b\rfloor\} - 8$. All budgets in the displayed calculation make these horizons positive. The common worst initialization is $|e| = r$, so observation-independent fixed mixtures cannot improve on the minimum of these lower bounds.

C. Paid continuation and the same-stream correction

The probe charges 288 messages and eighteen clock units. For commitment $a = (q, b)$ the applied horizon is

$$N_a = \min\{\lfloor (16B - 288)/(8+q)\rfloor, \lfloor (4B - 18)/b\rfloor\} - 8.$$

Let $u_{j,a} = (1 + N_a v_{j,+}(a))/(N_a+1)^2$, with $v_{j,+}$ evaluated at the upper variance and persistence endpoints, and set $\kappa_a = [\rho_+/((N_a+1)(1-\rho_+))]^2$, where $\rho_+ = (41/200)^b$. Then $J_{j,a}^r \leq u_{j,a}$ and $k_{j,a}^2 \leq \kappa_a$

uniformly. The initialization cross term vanishes because the frozen likelihood decision is even in the contrasts under every allowed centered Gaussian law. For the error event, $\mathbb{E}|m_j^2 - s_j^2| = 4\varphi(1)s_j^2 \le \theta_{j,+}$. The absolute continuing-state selection correction is therefore at most $\theta_{j,+}(\kappa_c + \kappa_w)$. With directional bound δ and commitments c_j, w_j, a rational upper bound is

$$U_j = (1-\delta)u_{j,c_j} + \delta \max\{u_{j,c_j}, u_{j,w_j}\} + \theta_{j,+}(\kappa_{c_j} + \kappa_{w_j}).$$

It follows from $J_c + p(J_w - J_c) \le (1-\delta)J_c + \delta \max(J_c, J_w)$ for $p \le \delta$. Use $\delta = 1/10$ for the full box and $\delta = 1/40$ with the exact reference parameters.

For a given prior π, let $L(\pi)$ be the minimum of the all-prefix lower bounds and $U(\pi) = (1-\pi)U_0 + \pi U_1$. The lower bound L is concave in π: each action's bound is a minimum of affine functions, and minimizing over actions preserves that form. The upper bound U is affine. If $U(\pi_i) \le (1-\epsilon)L(\pi_i)$ at both endpoints $\pi_i \in \{9/20, 11/20\}$, concavity proves the same inequality throughout the interval. Since $I_{\mathrm{fix}} \ge L$ and $R_{\mathrm{paid}} \le U$, this proves the uniform relative guarantee. Every number defining these bounds is rational.

D. Extending the comparison to predetermined resource cycles

Constant programs are included as period-one words. For a word $a_1, \dots, a_\ell$, $1 \le \ell \le 3$, put $\rho_t = \lambda^{b_t}$ and let z_t^* be the unique periodic solution of $z_t^* = \rho_t(1 + z_{t-1}^*)$. Existence follows because the full-cycle multiplier $\prod_t \rho_t < 1$; solving its scalar affine fixed point gives rational z_t^* at rational parameters. Write $\bar{u} = \ell^{-1}\sum_t q_t^{-1}$ and $\bar{z} = \ell^{-1}\sum_t z_t^*$. Define the cycle remainder for $0 \le s < \ell$ by

$$\epsilon_s = \sum_{t=1}^{s}(q_t^{-1} - \bar{u}) + 2\theta \sum_{t=1}^{s}(z_t^* - \bar{z}).$$

The finite-prefix covariance recurrence starts at $z_1 = 0$ and satisfies $z_t = \rho_t(1 + z_{t-1})$ thereafter. Thus $z_t = z_t^* - z_1^* \prod_{u=2}^{t} \rho_u$. Its accumulated noise variance obeys

$$\sum_{t=1}^{N}[q_t^{-1} + \theta(1 + 2z_t)] \ge Nv - C,$$

$$v = \bar{u} + \theta(1 + 2\bar{z}),$$

$$C = \frac{2\theta z_1^*}{1 - \max_t \rho_t} - \min_s \epsilon_s.$$

The bound uses a geometric upper bound on the missing initial-cycle covariance and retains every cycle phase. Replacing t by C in the preceding endpoint argument gives a lower bound over all prefixes of this cycle.

TABLE VI
EXACT-ARITHMETIC FINITE-BUDGET CERTIFICATES: CONSERVATIVE RELATIVE-RISK REDUCTIONS, NOT SAMPLED ESTIMATES. A DASH MEANS THESE BOUNDS CERTIFY NO POSITIVE REDUCTION, NOT THAT IMPROVEMENT IS IMPOSSIBLE. ALL FIVE DECLARED BUDGET SCALES ARE RETAINED.

Budget scale B	Reference model	Full parameter box
32	—	—
128	—	—
512	$\ge 2.65\%$	—
2048	$\ge 5.21\%$	$\ge 0.51\%$
8192	$\ge 5.81\%$	$\ge 1.15\%$

Let $m_s = \sum_{t=1}^{s}(8 + q_t)$ and $e_s = \sum_{t=1}^{s} b_t$, including $m_0 = e_0 = 0$. Define the number of affordable complete cycles before residue s by

$$C_s(B) = \min\left\{\left\lfloor \frac{16B - m_s}{m_\ell} \right\rfloor, \left\lfloor \frac{4B - e_s}{e_\ell} \right\rfloor\right\}.$$

The maximal scheduled prefix and applied horizon are exactly

$$M_{\max} = \max_{0 \le s < \ell}\{\ell C_s(B) + s\},$$

$$N_{\max} = M_{\max} - 8.$$

Every residue is feasible for the displayed budgets. Apply the lower bound to all 4368 words, using the lower parameter endpoints as before. The first-prefix lower bounds are positive for every word at both prior endpoints in this calculation. Each resulting lower bound remains a minimum of affine functions of the prior, so the same endpoint argument proves uniformity on $[0.45, 0.55]$. The fixed comparator is permitted to know the actual parameter pair and prior, but not the realized regime or initial error. Its observation-independent mixtures cannot improve on the minimum prefix risk because their worst initialization is the same scalar endpoint. The comparison is stronger than constant participation and spacing, while its declared period bound remains explicit.

The cyclic reference includes the alternating word $(4,2),(4,4)$, whose leading risk coefficient $205889/208320$ is smaller than the best constant coefficient $773/768$. Accounting for this stronger baseline reduces the reference guarantee at $B = 512$ from 4.17 to 2.65 percent. For the full box at $B = 8192$, the smaller endpoint guarantee is at $\pi = 11/20$. Rational outward rounding gives $L \ge 128741864/10^{12}$ and $U \le 127259404/10^{12}$. Cross multiplication shows $1 - U/L > 0.0115$. At $B = 2048$ the analogous endpoint bound exceeds 0.0051. Table VI separates these finite population guarantees from the empirical all-design primary comparison. The construction is an outcome-known analytical illustration of a declared

uncertainty box; it changes no completed study's decisions, estimands or validation status.

XXIV. Separation from Every Predetermined Resource Schedule

The comparison can cover arbitrary time variation without restricting a schedule's period. We first give a general lower bound for a running mean, then apply it to the paid rule in main Theorem 7.

A. A resource-envelope bound

Let $\mathcal{Q}, \mathcal{B} \subset \{1, 2, \ldots\}$ be finite participation and spacing sets, with overhead $h \geq 0$, message allowance M and environment allowance E. The common noise is stationary Gaussian AR(1), with variance $\theta > 0$ and persistence $0 \leq \lambda < 1$. Private observation noises are independent over time and have variance $1/q$ after averaging q actors. Let g_q be the decreasing piecewise-linear convex envelope through $(q, 1/q)$ for $q \in \mathcal{Q}$ and g_b the corresponding envelope through (b, λ^b) for $b \in \mathcal{B}$. Extend each envelope constantly to the right of its largest knot. Put $n_{\max} = \min\{M/(h + \min \mathcal{Q}), E/\min \mathcal{B}\}$. Assume $n_{\max} \geq 1$; otherwise stop is the sole feasible output. For $1 \leq n \leq n_{\max}$ define

$$v(n) = g_q(M/n - h) + \theta[1 + 2g_b(E/n)],$$
$$H(n) = \frac{r^2 + nv(n) - 2\theta\lambda}{(n+1)^2}.$$

Let $\mathcal{T}$ contain $1, n_{\max}$ and all points $M/(h+q)$ and E/b lying in $[1, n_{\max}]$.

Proposition 4 (Period-free resource envelope)**.** *Every predetermined schedule and affordable stopping prefix, using the running mean with initialization error $|e| \leq r$, has worst-initialization risk at least*

$$L(M, E; \theta, \lambda, r) = \min\{r^2, \min_{n \in \mathcal{T}} H(n)\}.$$

The same bound holds for observation-independent randomized schedules and stopping lengths. It also holds when terminal updates are withheld but charged.

To prove the bound, fix a deterministic applied prefix of length $N \geq 1$, with resource pairs (q_t, b_t) and observation times τ_t satisfying $\tau_t - \tau_{t-1} = b_t$ for $t \geq 2$. Its risk numerator at $|e| = r$ is exactly

$$r^2 + \sum_{t=1}^{N} q_t^{-1} + \theta N + 2\theta \sum_{s<t} \lambda^{\tau_t - \tau_s}.$$

All common covariances are nonnegative. Retaining adjacent pairs and adding back the first declared spacing gives

$$\sum_{s<t} \lambda^{\tau_t - \tau_s} \geq \sum_{t=2}^{N} \lambda^{b_t} \geq \sum_{t=1}^{N} \lambda^{b_t} - \lambda.$$

TABLE VII
Guaranteed risk reductions against every predetermined schedule. All quantities are analytical population bounds; a dash denotes no positive certificate from this envelope.

Budget scale B	Reference model	Full parameter box
32	—	—
128	—	—
512	$\geq 1.36\%$	—
2048	$\geq 4.89\%$	—
8192	$\geq 5.72\%$	$\geq 0.43\%$

For $\lambda = 0$ both sides are zero. The envelopes interpolate convex decreasing functions at their allowed actions, so Jensen's inequality and $\sum_t (h + q_t) \leq M$, $\sum_t b_t \leq E$ imply

$$\sum_t q_t^{-1} \geq N g_q(M/N - h), \quad \sum_t \lambda^{b_t} \geq N g_b(E/N).$$

These inequalities relax integer action counts and charge no extra cost to the fixed comparator. Their combination bounds its risk below by $H(N)$. Between consecutive points in $\mathcal{T}$, its numerator is $an + c$, where $a > 0$. The derivative of $(an+c)/(n+1)^2$ has numerator $a - an - 2c$, which decreases with n. It has no interior minimum; hence the minimum over all real $n \in [1, n_{\max}]$ occurs at a listed endpoint. Allowing real n can only strengthen the comparator relative to an integer schedule. Stop gives risk r^2. Every deterministic schedule's risk is increasing in e^2, so the common endpoints $e = \pm r$ also prove the randomized claim. Ignoring the cost of withheld updates enlarges the comparator's feasible set and preserves the lower bound.

The same argument applies to prior risk under model-specific persistence parameters by first lowering each nonnegative covariance entry to λ_- and replacing θ by $\bar{\theta}_- = \sum_j \pi_j \theta_{j,-}$. It does not presume that a predetermined schedule is constant, stationary or periodic.

B. Finite separation with the unchanged paid rule

Use $M = 16B$, $E = 4B$, $h = 8$, $r = 1$ and $\mathcal{Q} = \mathcal{B} = \{1, 2, 4, 8\}$. For the reference parameters, put $\theta = 41/40$, $\lambda = 1/5$. For the full box, put $\theta = (1 - \pi)99/2000 + \pi 99/50$, $\lambda = 39/200$. The paid upper bounds $U(\pi)$ from Section XXIII are unchanged: eighteen observations, both continuation horizons, eight withheld updates, the frozen detector and its uniform directional error bound remain fully charged. At each prior endpoint, evaluate L over $\mathcal{T}$ using rational arithmetic. It is concave in π, being a minimum of affine expressions, whereas U is affine. Thus endpoint inequalities $U \leq (1 - \epsilon)L$ imply the same relation for every prior between them.

At $B = 8192$, the smaller parameter-box endpoint reduction exceeds 0.0043192. Exact cross multiplication with the retained rational L and U gives the rounded guarantee 0.0043 in the main theorem. This bound resolves the comparison with all open-loop resource words in the catalogue, including arbitrarily long cycles, nonperiodic schedules, every prefix and independent random mixtures. The argument uses a continuous resource relaxation, not finite-word enumeration. The short-cycle calculation in Table VI remains a sharper certificate for that smaller comparison class. The general envelope and the paid upper bound separate the gain from knowing a covariance regime from the gain obtainable by scheduling without observing that regime.

XXV. Known-Model Population Risk of a Frozen Likelihood Commitment

The registered finite-seed result and the known-model population effect answer different questions. The latter can be evaluated without new trajectories for the unchanged binary likelihood rule, including its same-stream state dependence and all paid horizons. The distributional tool is classical: the positive chi-square mixture is recorded in Solomon, Section2, Eq.(2), with the Gaussian quadratic-form Laplace product in Eq.(3) [50]. The derivation below supplies its continuing-risk interface, not a new distribution theorem.

A. Diagonal likelihood event and state moment

Let $K_0 \prec K_1$ be the two positive definite contrast covariances, with contrast dimension $d = n - 1$ and balanced public prior. Write $K_0 = LL^\top$ and diagonalize

$$L^{-1}K_1L^{-\top} = V\,\mathrm{diag}(\ell_i)V^\top, \quad \ell_i > 1.$$

Under true model j, the coordinates of $V^\top L^{-1}Y$ have independent variances ℓ_i^j. With independent standard normal G_i, the likelihood rule selects the high model on

$$H = \left\{ \sum_i a_{j,i}G_i^2 > c \right\}, \quad c = \sum_i \log \ell_i,$$

where $a_{0,i} = 1-\ell_i^{-1}$ and $a_{1,i} = \ell_i - 1$. Both weight vectors are strictly positive; this positivity uses the ordered variance models and is not asserted for arbitrary covariance pairs. If $m_j(Y) = \sum_i b_{j,i}G_i$, then $s_j^2 = \sum_i b_{j,i}^2$. Let $p_j = \Pr_j(H)$ and $M_j = \mathbb{E}_j[m_j(Y)^2\mathbf{1}_H]$. Every cross-coordinate term vanishes by changing the sign of one G_i while leaving H unchanged. Consequently $M_j = \sum_i b_{j,i}^2\mathbb{E}[G_i^2\mathbf{1}_H]$.

For the two frozen post-probe programs, write $J_{j,0}^r, J_{j,1}^r$ for the exact paid stationary risks and k_0, k_1 for their common-state coefficients. The exact worst-initialization model risk is

$$R_j^{\mathrm{LR}} = J_{j,0}^r + (J_{j,1}^r - J_{j,0}^r)p_j + (k_1^2 - k_0^2)(M_j - s_j^2p_j). \tag{S33}$$

The odd initialization term integrates to zero because H is even. All horizons and coefficients in Eq. (S33) are those remaining after payment; there is no independent learning reset. The criterion $\sum_j \pi_j R_j^{\mathrm{LR}} < I_{\mathrm{full}}$ is therefore an exact known-model comparison for this fixed paid two-commitment rule against the complete fixed-prefix benchmark. It does not optimize over different probes or all measurable selectors.

B. Positive mixture and explicit series remainder

Suppress j, let $\beta = \min_i a_i > 0$, $r_i = 1 - \beta/a_i \in [0, 1)$, and define

$$C(z) = \prod_i (\beta/a_i)^{1/2}(1 - r_iz)^{-1/2} = \sum_{k\geq 0} c_kz^k.$$

The coefficients are nonnegative and $C(1) = 1$. Logarithmic differentiation gives

$$c_0 = \prod_i (\beta/a_i)^{1/2}, \quad c_k = \frac{1}{k}\sum_{l=1}^{k}\left(\frac{1}{2}\sum_i r_i^l\right)c_{k-l}.$$

Substitute $z = (1 + 2\beta t)^{-1}$ into the Laplace product $\prod_i(1 + 2a_it)^{-1/2}$. It becomes a probability mixture of the transforms of $\beta\chi^2_{d+2k}$, proving the classical series. Define h_k by the nonnegative generating function

$$\sum_{k\geq 0} h_kz^k = C(z)\sum_i \frac{b_i^2\beta/a_i}{1 - r_iz}.$$

Its coefficient mass is $\sum_k h_k = s^2$. Weighting one Gaussian coordinate by G_i^2 increases that coordinate's chi-square degrees of freedom from one to three. Thus, with $\overline{F}_v$ the standard chi-square upper-tail function,

$$p = \sum_{k\geq 0} c_k\overline{F}_{d+2k}(c/\beta),$$
$$M = \sum_{k\geq 0} h_k\overline{F}_{d+2+2k}(c/\beta).$$

For sums retained through K, denote the partial values by p_K, M_K and let $\epsilon_p = 1 - \sum_{k\leq K} c_k$, $\epsilon_M = s^2 - \sum_{k\leq K} h_k$. In exact arithmetic,

$$p_K \leq p \leq p_K + \epsilon_p, \quad M_K \leq M \leq M_K + \epsilon_M.$$

The corresponding risk interval width is at most

$$|\Delta J_j - \Delta k^2 s_j^2|\epsilon_p + |\Delta k^2|\epsilon_M,$$
$$\Delta J_j = J_{j,1}^r - J_{j,0}^r, \quad \Delta k^2 = k_1^2 - k_0^2.$$

Floating-point rounding and special-function error are separate from this exact series-truncation statement.

C. Outcome-known diagnosis of the unchanged gate

Deterministic numerical evaluation retains all128 observable-study designs and their original29 admissions and99 fixed fallbacks. It approximates the population normalized fixed-minus-gate gain by 0.00875334, below the registered 0.01 practical threshold, and the uncertified-probe-minus-gate gain by 0.190124. The maximum normalized omitted-series-mass risk width at the computed coefficients is below 3.7×10^{-14}; this is a truncation diagnostic, not a rigorously rounded interval or a sampling confidence limit. Scaled chi-square identities and independent two-dimensional quadrature check the probability and selected-state-moment formulas. Independent characteristic-function integration checks the actual probe laws separately. This outcome-known calculation neither changes an action nor replaces the registered negative finite-seed decision. It explains why simply adding seeds is not a principled response to the unmet practical-size target on this grid.

XXVI. A Matrix Form of Paid Calibration Value

The scalar initialization argument relies on the fact that every program contracts all error directions by the same scalar. For vector learners this property can fail, so replacing every program's initial error by its separate worst direction can produce an unattainable score. The following reduction keeps one common unknown initialization across the whole observation space. Its minimax interchange is a classical convex-game argument; the specific object is the finite-dimensional covariance problem for paid continuing programs.

A. Information, affine continuation and integrability

Let $e \in \mathbb{R}^d$ satisfy $e^\top P e \le r^2$, with public $P \succ 0$ and $r > 0$. The unobserved e is fixed across the experiment and does not affect the centered calibration-probe law. Fix a public prior over a finite set of Gaussian noise models j. For each model, the probe law $Y \sim \mathcal{N}(0, \Sigma_j)$ is nonsingular, the conditional common state $C \in \mathbb{R}^k$ has linear mean $m_j(y)$ and constant covariance, and the continuing Gaussian noise is jointly specified. Both the paid probe schedule and every subsequent candidate program are frozen before observing Y. Under model j and program a, suppose the terminal error has conditional mean $A_{j,a}e + B_{j,a}m_j(y)$ and conditional covariance $V_{j,a} \succeq 0$. All matrices are known and deterministic, and their horizons include the probe charges, affine delay and withheld updates. An implementable selector sees Y and these public objects but not e or j. In particular the centered calibration observation must be obtainable without subtracting an unknown target.

Let g be the mixture density and $p_j(y)$ its model posterior, and define

$$H_a(y) = \sum_j p_j(y) A_{j,a}^\top A_{j,a},$$
$$Q_a(y) = \sum_j p_j(y)\{\operatorname{tr} V_{j,a} + \|B_{j,a} m_j(y)\|^2\}.$$

Both functions are even in y, H_a is bounded, and Q_a is integrable under g because the catalogue is finite and Gaussian second moments are finite. The posterior squared loss equals $e^\top H_a(y) e + Q_a(y)$ plus a term linear in e and odd in y. Consequently symmetrizing a selector cannot increase its maximum risk over the centrally symmetric error set.

B. Exact covariance reduction

Let Δ_{even} be the measurable randomized even selectors over the finite catalogue, and put

$$\mathcal{S} = \{S \succeq 0 : \operatorname{tr}(PS) \le r^2\}.$$

Then the exact robust risk following the paid probe is

$$\inf_\delta \sup_{e^\top P e \le r^2} R_e(\delta)$$
$$= \max_{S \in \mathcal{S}} \int \min_a \{\operatorname{tr}[H_a(y) S] + Q_a(y)\} g(y)\, dy. \quad \text{(S34)}$$

The minimax error-direction problem has therefore reduced to a concave maximization over a finite-dimensional initialization covariance. The probe integration is still present; its numerical approximation requires its own accuracy justification.

To prove Eq. (S34), first symmetrize any selector, using $R_e(\delta^{\mathrm{sym}}) = [R_e(\delta) + R_{-e}(\delta)]/2$. For an even selector define $M_\delta = \mathbb{E}_g \sum_a \delta_a(Y) H_a(Y)$ and $c_\delta = \mathbb{E}_g \sum_a \delta_a(Y) Q_a(Y)$. Its risk is $e^\top M_\delta e + c_\delta$. The convex hull of $ee^\top$ over $e^\top P e \le r^2$ is exactly $\mathcal{S}$: diagonalize $P^{1/2} S P^{1/2}$ and express it as a convex combination of rank-one radius-r errors, with any unused mass at zero. Hence the worst-case risk equals $\max_{S \in \mathcal{S}} \operatorname{tr}(M_\delta S) + c_\delta$.

The set $\mathcal{S}$ is compact and convex. Regard each component of δ as an element of $L^\infty(g)$. The probability-simplex constraints and evenness are weak-star closed, so Δ_{even} is weak-star compact and convex. The integrable functions H_a, Q_a make the objective continuous and affine in each argument for the weak-star and finite-dimensional topologies. The classical minimax interchange applies. At fixed S, an even first minimizer of the finitely many even scores minimizes the integral, proving Eq. (S34). Continuity and compactness give attainment in the primal and covariance-dual problems.

C. A dual maximizer does not license arbitrary tie breaking

There exists a saddle pair (δ^*, S^*); an arbitrary pointwise minimizer at a dual maximizer S^* need not be a minimax selector. Its tie randomization must also make S^* a worst-case covariance for that selector. For example take an irrelevant nonsingular centered Gaussian probe, zero continuation noise, $P = I_2$, $r = 1$ and two programs with $A_0 = \mathrm{diag}(1, 0)$ and $A_1 = \mathrm{diag}(0, 1)$. The dual objective is $\min(S_{11}, S_{22})$, maximized at $S^* = I_2/2$ with value $1/2$. Both actions tie everywhere at S^*. Selecting the first action deterministically has worst-case risk one; mixing the two actions equally attains $1/2$. Thus the scalar fixed tie rule cannot be transported to arbitrary matrix contractions without checking the saddle condition. If the minimizing action is unique g-almost everywhere at a dual maximizer, the unique pointwise minimizer is the saddle selector because a primal optimizer exists and must minimize there.

D. The strongest fixed comparison is itself a semidefinite program

For the full-budget no-probe catalogue let $\overline{H}_a^{\mathrm{full}}$ be the prior average of $A_{j,a}^\top A_{j,a}$ and $\overline{Q}_a^{\mathrm{full}}$ its full-budget zero-initialization risk. A fixed randomized program uses a probability vector $w \in \mathcal{W} = \{w \geq 0 : \sum_a w_a = 1\}$ independent of the calibration observations and unknown initialization. Its optimum worst-case risk I_{fix} is the value of

$$\begin{aligned} \underset{w \in \mathcal{W},\, t \in \mathbb{R}}{\text{minimize}} \quad & r^2 t + \sum_a w_a \overline{Q}_a^{\mathrm{full}} \\ \text{subject to} \quad & \sum_a w_a \overline{H}_a^{\mathrm{full}} \preceq tP. \end{aligned} \tag{S35}$$

The constraint bounds the largest generalized initialization eigenvalue exactly, so this is a semidefinite program over a finite catalogue. The example above proves why a comparison restricted to deterministic fixed programs can overstate a matrix selector's benefit. Strict superiority to every fixed randomized programme in this catalogue is attainable after the specified paid probe if and only if the value in Eq. (S34) is strictly less than I_{fix}. The reverse inequality is an impossibility certificate for this probe and catalogue, rather than a statement about every possible adaptive experiment.

With $P = I$ and scalar contractions $A_{j,a} = d_a I$, every unknown direction gives the same coefficient and the maximizing covariance can be taken on the radius boundary. The formula reduces to the previously proved scalar even conditional-risk selector, and a fixed random mixture cannot improve its linear scalar endpoint objective. The scalar simulations evaluate this reduction; the matrix characterization is analytical.

XXVII. Conditional-State Validation and the Paid-Probe Boundary

The conditional-state study fixes nine policies, two learners, sixteen families, four budget scales, two covariance regimes, both initial-error signs and 64 independent paired master seeds. Its 294,912 trajectories use a separate known-zero calibration pilot, rather than unknown-target contrasts. Public control rules see neither the realized regime, true target, initial-error sign nor future observations. The two evaluation-only oracles are labeled separately. The post-probe catalogue contains stop and every original participation–spacing program at its maximal affordable horizon; the full-budget fixed benchmark uses the corresponding full-budget catalogue and stop. The fixed-minus-conditional and posterior-marginal-minus-conditional contrasts are averaged equally over all 128 designs, both regimes and both signs within each master seed, normalized by each design's pre-data population best balanced fixed risk. The registered positive decision requires two positive one-sided 99.5 percent paired-bootstrap lower bounds and at least one percent mean fixed-policy improvement. The bootstrap uses 10,000 resamples and approximate percentile coverage.

The registered decision is negative. The fixed-minus-conditional mean is -0.993855, with descriptive 95 percent interval $[-1.015340, -0.973059]$. The posterior-marginal-minus-conditional mean is -1.12×10^{-5}, with interval $[-1.06 \times 10^{-4}, 5.67 \times 10^{-5}]$. There are 87 adverse fixed-comparison design means out of 128. Fixed-minus-conditional means at scales 32, 128, 512 and 2048 are respectively -3.728961, -0.237679, -0.052351 and 0.043571. Thus selecting a continuing program after paying a feasible pilot does not replace the economic decision of whether to buy that pilot. The state-information correction gives no confirmed incremental gain over the posterior-marginal rule on this grid.

XXVIII. Observable Paid-Risk Gate Against All Fixed Prefixes

A. Prospective model, information and baseline

The observable-probe study was specified before the conditional-state study's outcomes were inspected. It retains nine policies, two learners, all sixteen families and four budget scales, both true covariance regimes and both hidden initial-error signs. There are 64 fresh paired master seeds, $20591001, \ldots, 20591064$, giving 294912 policy trajectories and the same number of full semantic replays. The known binary models, balanced public prior, radius $r = 1$, original participation–spacing sets, learner steps and paid FIFO convention are unchanged. The raw probe is target contaminated: it observes target plus noise for the running mean, and half the target plus

TABLE VIII
COMPLETE CONDITIONAL-STATE STUDY: DESCRIPTIVE EQUAL-CELL MEANS OVER ALL FAMILIES, BUDGETS, REGIMES AND INITIALIZATION SIGNS.

Policy	Squared risk	Normalized risk	Messages	Clock units
Robust conditional	0.098475	2.009532	18509.68	2468.19
Known-error oracle (evaluation only)	0.098169	2.009318	18509.47	2466.34
Posterior marginal	0.098476	2.009521	18509.70	2468.19
Legacy risk gate	0.077053	1.270375	18510.16	2463.49
Information only	0.098477	2.009903	18509.93	2467.45
All actors	0.113417	2.962129	19040.00	680.00
Fixed $(q, b) = (2, 1)$	0.061440	1.326310	19037.50	1903.75
Best balanced full-budget fixed	0.051784	1.015677	18187.84	2560.36
Known-regime fixed (evaluation only)	0.051252	0.987827	18508.48	2509.63

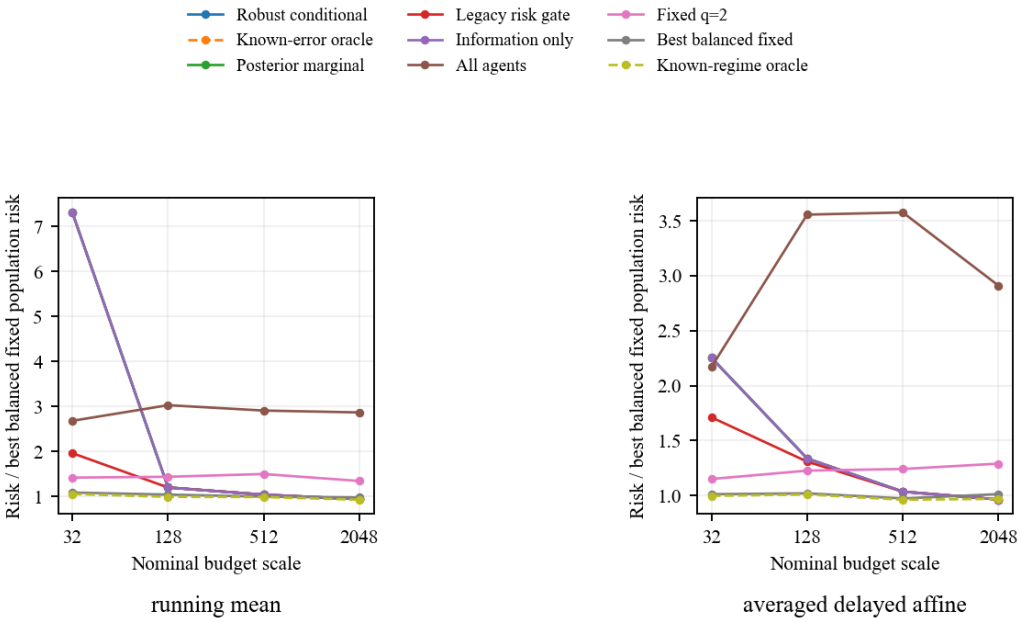


Fig. 9. Conditional-state study, all nine policies: normalized risk against budget scale, separately for the two learners. Coincident curves are not graphically offset.

noise for the affine learner at initial parameter zero. The deployment uses only first differences and the equivalent two-filter invariant likelihood, not the unknown target, initial-error sign, true regime or future observations. Both target signs use the same potential noise stream across policies; the parameter and target are translated together.

The probe uses all Q actors at unit spacing and the smallest length $n \in \{2, \dots, 4096\}$ whose projected Gaussian affinity is at most 0.025. The two post-probe commitments minimize the appropriate model's exact risk over stop and every original participation–spacing pair at its maximal affordable remaining horizon. The signed prior-risk upper bound is compared with the best balanced full-budget fixed risk before collecting any probe observations. Unlike the preceding study, that fixed benchmark includes every affordable scheduled training prefix for every original pair, plus stop, at the same learner and step. An independent scalar recurrence checks its population normalizer; evaluation-only known-regime fixed programs receive the same complete prefix catalogue. No choice of learning-rate sequence or estimator outside the declared class is included.

The certified gate probes only on a strict certificate improvement and otherwise executes that strongest fixed prefix without a probe. Other policies are an uncertified invariant likelihood commitment whenever probing is feasible, a no-carryover admission ablation, posterior-marginal selection over the full post-probe catalogue, all actors, fixed $(2, 1)$, and two privileged evaluation

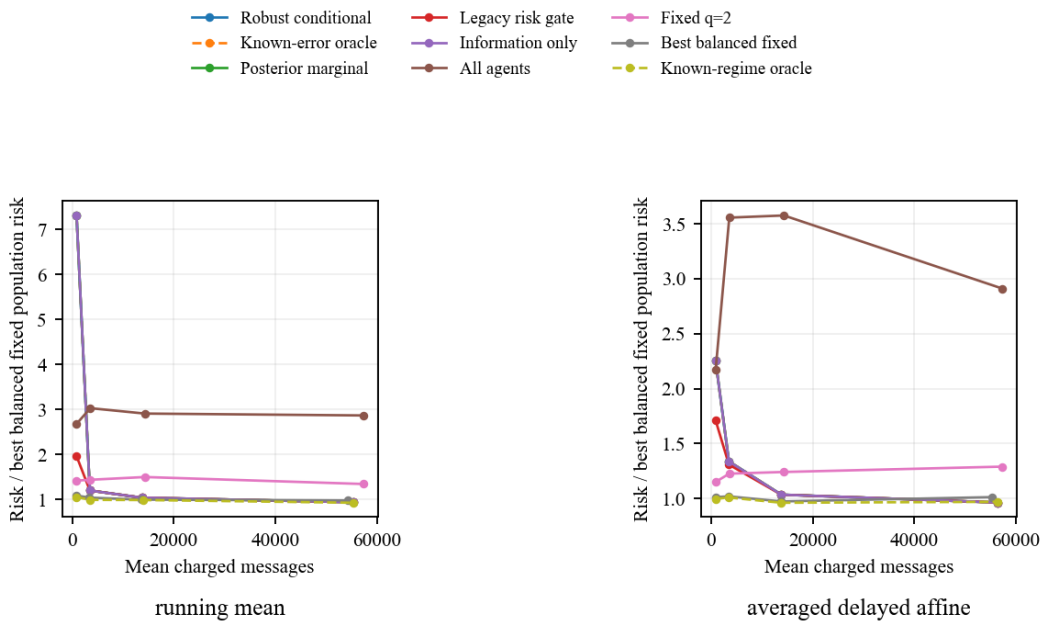


Fig. 10. All conditional-state policies against actual paid messages; points correspond to distinct budget experiments.

references. The known-target centered-pilot reference and known-regime fixed reference are not deployable policies or universal risk lower bounds. All probe and scheduled training observations are paid, including withheld terminal updates. An infeasible probe uses the strongest fixed backstop.

B. Registered primary decision and every policy

The paired master-seed estimands normalize each design by its pre-data balanced population best-fixed risk and weight all 128 designs, both regimes and both signs equally. The two co-primary contrasts are fixed minus certified gate, and uncertified invariant likelihood probe minus certified gate. The registered positive decision requires both one-sided 99.5 percent bootstrap lower bounds above zero and fixed-minus-gate mean at least 0.01. There are 10,000 fixed-seed bootstrap resamples; coverage is approximate percentile coverage, not a finite-sample confidence theorem.

The registered practical-positive decision is not confirmed. Fixed-minus-gate mean is 0.00444303, with one-sided lower bound -0.00317725 and descriptive 95 percent interval $[-0.00119855, 0.00999250]$. Uncertified-probe-minus-gate mean is 0.170210, with lower bound 0.148124 and descriptive interval $[0.153487, 0.187211]$. The former misses both its confidence and practical-size conditions; the latter supports the value of not buying an uneconomic probe. There are 9/128 adverse fixed-comparison design means and 20/128 adverse uncertified-probe comparison means. At scales 32, 128, 512 and 2048 the fixed-minus-gate means are respectively 0, 0, 0.00989743 and 0.00787469. The running-mean aggregate is 0.00876160 and the affine aggregate 0.000124462. These secondary descriptions do not replace the all-design primary decision.

There are 116 feasible probes, 29 certificate admissions and 99 exact fixed-prefix fallbacks. All 25,344 nonadmitted policy rows have exactly the paired fixed policy's terminal error and paid costs. Corrected and no-carryover gates coincide on all 32,768 paired rows; there is again no evidence of incremental empirical decision value for the correction on this grid. The two endpoint signs check information roles; uniform initialization risk control comes from the analytical envelope, not empirical coverage of every initialization.

The nearly equal raw pooled risks do not favor the gate over strongest fixed: 0.0506443 versus 0.0506405. Raw pooled and design-normalized contrasts give different weights to heterogeneous designs, so neither is silently substituted for the registered estimand. The gate sends more messages but uses slightly fewer clock units than strongest fixed; no equal-realized-cost dominance follows. The full mechanism diagnosis is therefore bounded: observable risk admission avoids the losses of paying a feasible pilot indiscriminately, but the study does not demonstrate a practically necessary advantage over a well-chosen fixed prefix. No extra seeds, favorable-subset replacement or outcome-

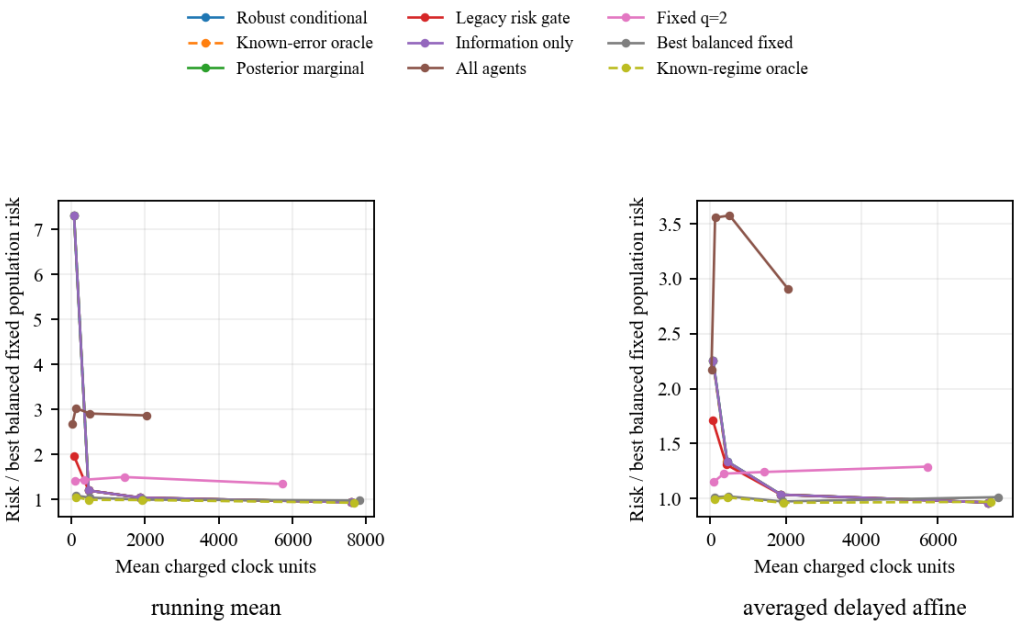


Fig. 11. All conditional-state policies against paid clock units; neither the message nor clock plot asserts equal-cost pointwise dominance.

TABLE IX
OBSERVABLE PAID-GATE STUDY: ALL NINE POLICIES, DESCRIPTIVE EQUAL-CELL MEANS OVER 512 LEARNER–FAMILY–BUDGET–REGIME–SIGN CELLS PER POLICY, WITH 64 SEEDS PER CELL.

Policy	Squared risk	Normalized risk	Messages	Clock units
Observable certified gate	0.050644	0.972057	18461.16	2516.16
Known-target pilot (evaluation only)	0.067765	1.141980	18504.42	2470.74
Observable uncertified likelihood	0.067856	1.142267	18506.78	2472.50
No-carryover admission	0.050644	0.972057	18461.16	2516.16
Observable posterior marginal	0.067856	1.142201	18505.12	2472.96
All actors	0.113505	2.941037	19040.00	680.00
Fixed $(q, b) = (2, 1)$	0.060679	1.346523	19037.50	1903.75
Best balanced full-budget fixed prefix	0.050640	0.976501	18187.84	2560.36
Known-regime fixed (evaluation only)	0.050115	0.949517	18508.48	2509.63

dependent threshold change is used to alter that decision.

APPENDIX A
PROOF OF THE AFFINE DELAYED MARKOV BOUND

Fix a stationary action $a = (q, b, \eta)$ and condition on the history immediately before the retained joint observation. All expectations below include the Markov observations; the delay profile is deterministic as specified in Assumption 1. Write $\bar{H}_k = q^{-1}\sum_i H_{i,k}$, $\bar{\xi}_k = q^{-1}\sum_i \xi_{i,k}$, and

$$v_k = e_k - \eta(\bar{H}_k e_k - \bar{\xi}_k), \quad r_k = \frac{1}{q}\sum_i H_{i,k}(e_{k-\tau_i} - e_k). \tag{S36}$$

Then Eq. (4) is $e_{k+1} = v_k - \eta r_k$.

Total-variation duality and stationary centering give

$$\left\|\mathbb{E}[\bar{\xi}_k \mid \mathcal{F}_k]\right\| \le 2G\delta. \tag{S37}$$

Moreover, stationarity and the bounded integrand imply

$$\mathbb{E}[\|\bar{H}_k e - \bar{\xi}_k\|^2 \mid \mathcal{F}_k] \le (2K_q + 4L^2\delta)\|e\|^2 + 2\Omega_q + 4G^2\delta. \tag{S38}$$

Combining Eq. (S37) with the strong monotonicity of A and applying Young's inequality with parameter μ_δ yields

$$\mathbb{E}\|v_k\|^2 \le a_\delta(\eta)R_k + \beta_\delta(\eta). \tag{S39}$$

The affine telescoping identity retains the innovation:

$$e_{k-\tau_i} - e_k = \eta \sum_{s=k-\tau_i}^{k-1} \left[\frac{1}{q}\sum_j H_{j,s} e_{s-\tau_j} - \bar{\xi}_s\right]. \tag{S40}$$

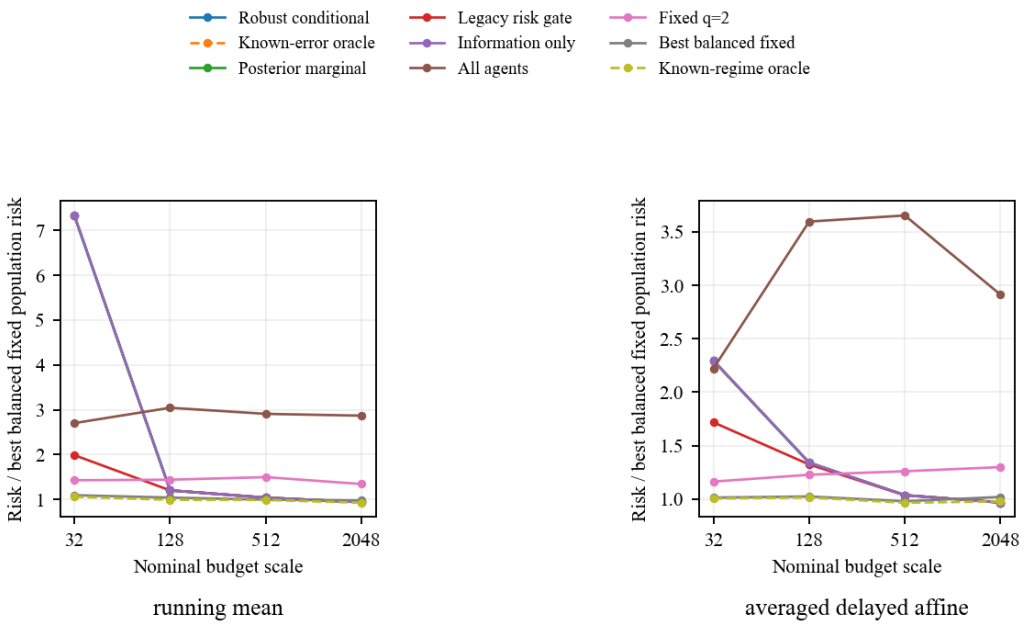


Fig. 12. Conditional-state study using the larger empirical mean over the two fixed initial-error signs in each learner and budget.

Jensen's inequality and the almost-sure bounds in Assumption 1 give

$$\mathbb{E}\|r_k\|^2 \le 2\eta^2 L^4 \tau_{\mathrm{rms}}^2 R_k + 2\eta^2 L^2 G^2 \tau_{\mathrm{rms}}^2. \tag{S41}$$

For any $\lambda > 0$,

$$\|v_k - \eta r_k\|^2 \le (1+\lambda)\|v_k\|^2 + (1+\lambda^{-1})\eta^2 \|r_k\|^2. \tag{S42}$$

Substituting Eq. (S39) and Eq. (S41) gives $\mathbb{E}\|e_{k+1}\|^2 \le c_\lambda R_k + d_\lambda$. The choice $\lambda = \sqrt{h_\tau/a_\delta}$ minimizes the contraction coefficient and produces Eq. (17). For zero delay, $r_k = 0$ and the continuous limiting definitions follow directly.

Let $R_\star = d_{\mathrm{aff}}/(1-c_{\mathrm{aff}})$. If $R_k \ge R_\star$, every new value entering the sliding envelope is at most $c_{\mathrm{aff}} R_k + d_{\mathrm{aff}} \le R_k$. If $R_k < R_\star$, the same one-step bound keeps every new value below $R_\star$. Thus, in both cases, each new value is at most $R_\star + c_{\mathrm{aff}}(R_k - R_\star)_+$. After $2D+1$ updates the entire old window is replaced, giving $(R_{k+2D+1} - R_\star)_+ \le c_{\mathrm{aff}}(R_k - R_\star)_+$. Iteration gives $R_{nm} \le R_\star + c_{\mathrm{aff}}^n (R_0 - R_\star)_+ \le c_{\mathrm{aff}}^n R_0 + R_\star$, proving Theorem 1.

A. *Proof of Corollary 3*

The shared-Young-parameter comparison and explicit moment-slack bounds are given in the supplementary material.

B. *Return transfer and arbitrary terminal indices*

Let $n = \lfloor N_a/(2D+1) \rfloor$. The envelope argument keeps every later incomplete-block iterate below $\max\{R_{n(2D+1)}, R_\star\}$, which is the truncated-transient form in Eq. (22). Since $J_\nu(w) - J_\nu(w^\star) = \psi_\nu^\top e$ for $\psi_\nu = \mathbb{E}_{s\sim\nu}\phi(s)$, Cauchy–Schwarz and the terminal envelope prove Corollary 1.

C. *Proof of Theorem 2*

The complete positive-part tracking and advantage-transfer proof is retained in the supplementary material.

D. *Existence and complexity of the mixed controller*

For fixed (q, b), every coefficient and Eq. (22) is continuous on the declared positive-step domain. The nonstrict certification margins make that domain closed and bounded, so the extreme-value theorem applies whenever it is nonempty. Finite enumeration across nonempty pairs gives a global minimizer. A numerical implementation attains this oracle minimum when its scalar solver has certified global accuracy. The $N_q N_b$ scalar searches cost $O(N_q N_b L_\eta)$ evaluations. For rank-one $H_{i,k}$, feature inner products avoid dense matrices in update aggregation, which costs $O(qd)$ arithmetic and $O(d)$ memory. Exact spectral moment calibration is charged separately from rank-one update aggregation.

APPENDIX B
CORRELATION-LIMITED MINIMAX RISK

For one round of the Gaussian location subclass, the agent average has variance $\sigma_c^2 + \sigma_e^2/q$. The T round averages are independent and Gaussian with known variance, so their sample mean is minimax for squared loss over the unrestricted location family and has risk $T^{-1}(\sigma_c^2 + \sigma_e^2/q)$. Dividing the single-agent risk by this value gives Eq. (10). Since $1+(q-1)\rho \geq q\rho$ for $\rho > 0$, the speedup is at most ρ^{-1}.

APPENDIX C
PROOF OF THE STOPPED ADAPTIVE INFORMATION IDENTITY

Condition on the pre-action history and realized action A_k. An orthogonal rotation maps X_k to $Y_k = q_k^{-1/2}\mathbf{1}^\top X_k$ and hypothesis-invariant standard-normal contrasts. Because the predictable action kernels cancel, the stopped log likelihood is the sum of scalar Kalman innovation ratios, where $Y_k \sim \mathcal{N}(\sqrt{q_k}m_{j,k}^-, V_{j,k})$. The conditional Gaussian divergence and the bounded-stopping chain rule give Eq. (26) and Eq. (29); exchanging hypotheses gives the reverse identity.

For a decision $\widehat{J}$, set $p_v = \mathbb{P}_v\{\widehat{J} = j\}$. Data processing gives

$$\mathrm{KL}(P_j^\pi|\mathcal{F}_\tau \,\|\, P_{1-j}^\pi|\mathcal{F}_\tau) \geq \mathrm{kl}(p_j, p_{1-j}). \tag{S43}$$

If both directional errors are at most δ, monotonicity of binary relative entropy yields Eq. (30).

APPENDIX D
OCCUPATION-MEASURE OPPORTUNITY LOWER BOUND

The supplementary material retains the complete stopped-flow proof of Proposition 4, including the occupation measure, information constraints and exact opportunity-loss identity.

APPENDIX E
PROOF OF THE MAIN LEARNING-VALUE GUARANTEE

Let $\mathcal{H}_d$ be the complete probe history. The declared conditional certificate gives $\mathbb{E}_j[\|e_{\mathrm{LA}}\|^2 \mid \mathcal{H}_d] \leq \overline{R}_j(\widetilde{a}_{\widehat{J}}; B_d^D, D)$ for either selected action. The right side is at most $\overline{R}_j^{\mathrm{sel}} + \mathbf{1}\{\widehat{J} \neq j\}W_j(d)$. Taking expectation and using the directional error bound proves Eq. (37).

Subtracting the full-budget all-agent certificate and using Eq. (34) gives

$$\mathbb{E}_1\|e_{\mathrm{LA}}\|^2 - \overline{R}_1^{\mathrm{all}}(B, D) \leq C_1^{\mathrm{probe}} + C_1^D - G_1 + \delta W_1 < 0. \tag{S44}$$

The same decomposition on instance zero is bounded above by $\epsilon_{\mathrm{safe}}(d) \leq \bar{\epsilon}$. The fallback statement follows directly from the strict admission rule.

APPENDIX F
PROOF OF THE THRESHOLD SANDWICH

The supplementary material gives the complete expected-information and scale calculation. Its lower bound uses uniform second moments of both Gaussian filters, not a pointwise bound on their conditional means. Combining that bound with Eq. (30) gives $B_N \geq c_N\,\mathrm{kl}(1-\delta, \delta)$. A fixed probe needs $n_\delta = O(\log(1/\delta)) = O(B_N + 1)$ observations. The risk expansion makes its high-instance benefit order s^{-1} and its lost-learning cost order $(n_\delta + 1)s^{-2}$. The assumed leading gap dominates the wrong-decision term; low-instance safety follows above the stated allowance scale. Taking the maximum of these sufficient scales proves Eq. (46).

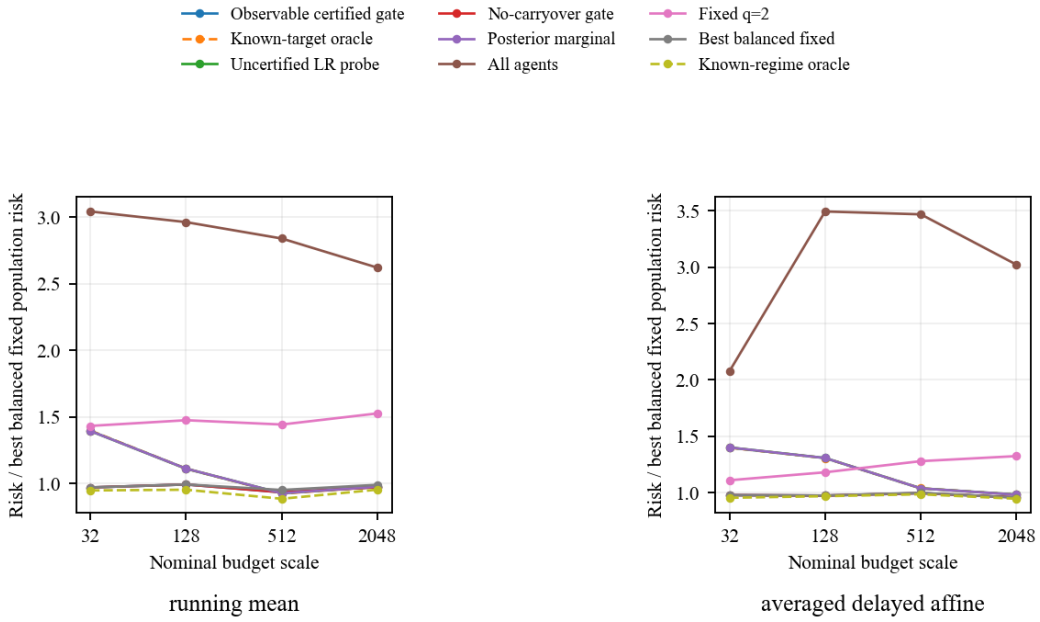


Fig. 13. Observable study, all nine policies against budget scale. The native legend's known-target oracle is an evaluation-only centered-pilot reference, not a universal optimum; coincident curves are retained without offsets.

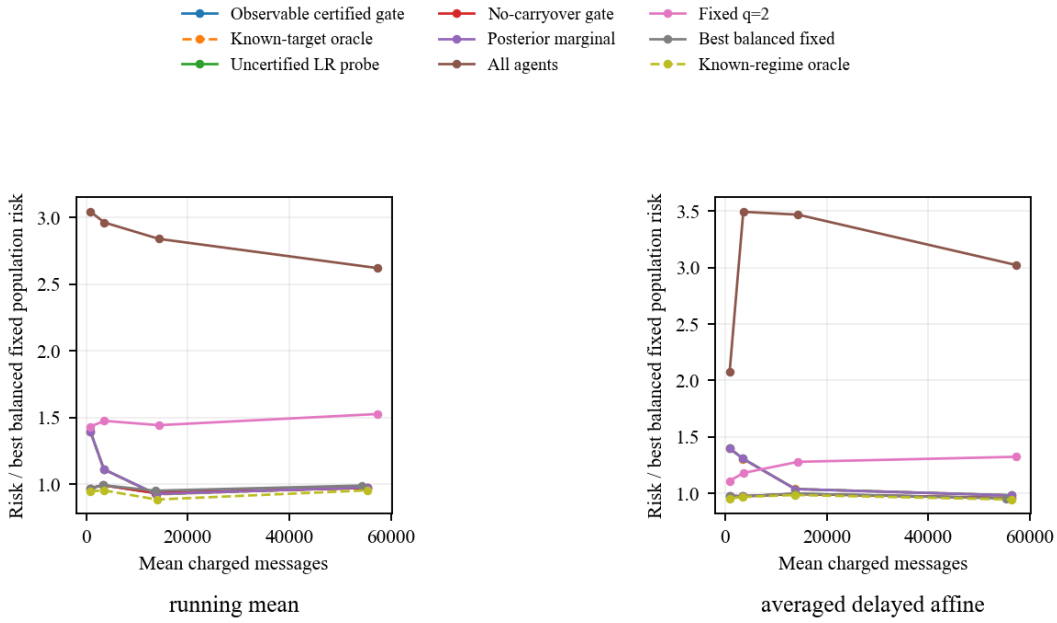


Fig. 14. All observable-study policies against actual paid messages. Points connect distinct budget experiments rather than a common-resource stopping comparison.

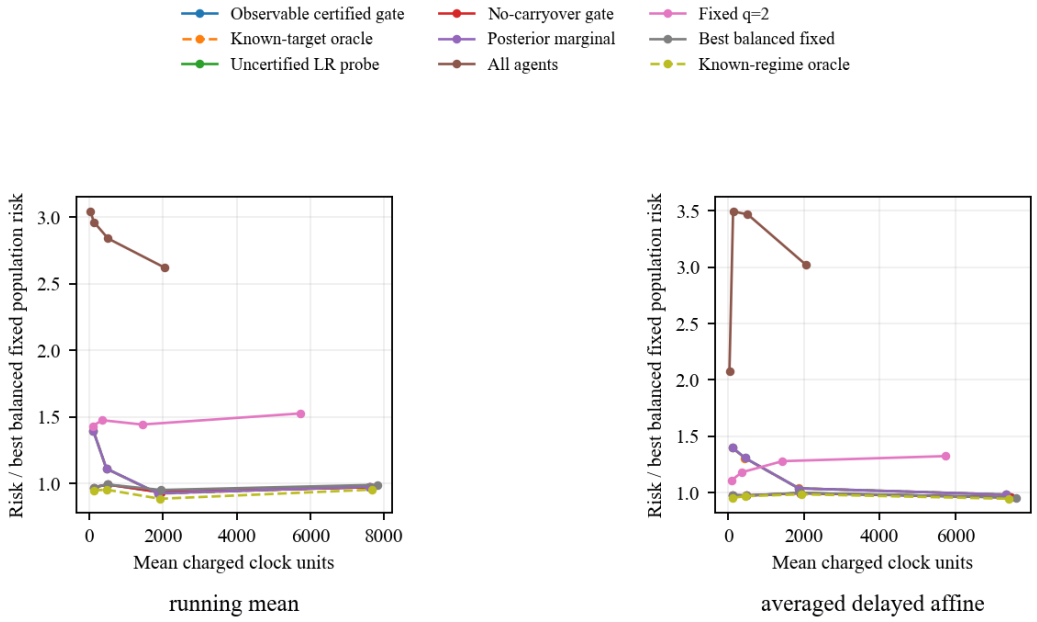


Fig. 15. All observable-study policies against paid clock units; the fixed-prefix backstop and nine adverse fixed-comparison design means remain in the analysis.

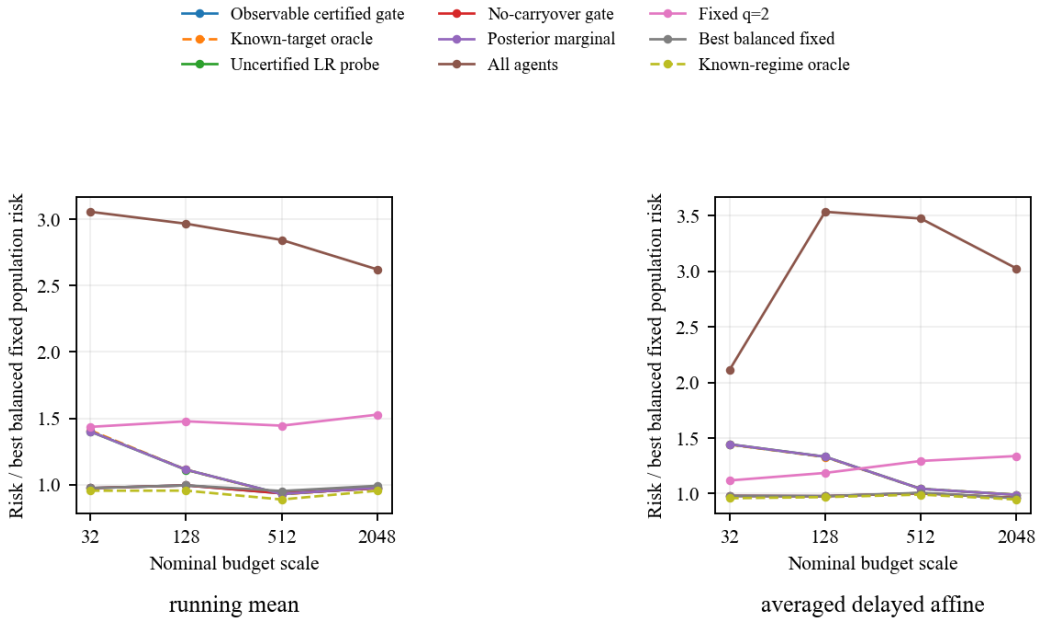


Fig. 16. Observable study using the larger empirical mean over the two fixed hidden signs for each learner and budget. This descriptive maximum is not a uniform confidence guarantee.